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DISCRETE STRUCTURE TENSORS

A MODERN REFORMULATION
OF HYPERSPATIAL GEOMETRY

Abstract

This monograph reconstructs the hyperspatial geometry of elementary matter on a fully discrete substrate. The cellular complex carries an exterior calculus with a Hodge star, a vielbein, a connection cochain and a curvature 2-form; from these data the dynamical equations of the theory are obtained by a deterministic metrisation procedure. The present volume develops the discrete-geometric machinery and applies it to the description of structure tensors, polymetric composition and the spin content of elementary cells.

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1

The Macroscopic Background of Microscopic Processes

Scope of the chapter. This chapter assembles the empirical and conceptual prerequisites on which the entire theory rests. We do not re-derive textbook physics—conservation laws, Lorentz invariance, Maxwell electrodynamics and the classical Einstein field equations are cited and used. Three genuinely non-standard proposals are introduced and developed in some detail:

1. a heuristic gravitomagnetic extension of Newtonian gravity, giving an antisymmetric four-tensor G_{km} analogous to the electromagnetic F_{km} ;
2. a composite real four-dimensional spacetime $R_4 = R_{+4} \otimes R_{-4}$ obtained by entangling a “gravito-temporal” auxiliary space R_{+4} with the Minkowski space R_{-4} of special relativity;
3. a non-Hermitian metric three-block decomposition $g_{ik} = g_{ik}^{(1)} + g_{ik}^{(2)} + g_{ik}^{(3)}$ leading to a non-Hermitian extension of Einstein’s field equation, which—once the quantum action principle is imposed—becomes a discrete relation $R_{ik} \propto w \eta_{ik}$ between geometric and combinatorial data.

The presentation deliberately keeps every Heim-specific proposition typographically distinguishable from textbook material so that the discrete-geometric machinery of Chapter 3 can be applied to it cleanly.

1.1 Empirical Foundations and Definitions

1.1.1 Empirical postulates

We adopt the following empirically well-founded principles as the inductive basis of the theory.

Principle 1.1 (Conservation). Energy, momentum, electric charge, and (per channel) baryon and lepton number are conserved under closed dynamics ([Wikipedia: Noether’s theorem](#)).

Principle 1.2 (Variational structure). The field equations of every macroscopic interaction follow from a stationary-action principle; in the same regime the entropy law of classical statistical mechanics holds ([Wikipedia: Principle of least action](#), [Wikipedia: Second law of thermodynamics](#)).

Principle 1.3 (Quantum of action). Every physical action is an integer multiple of Planck’s quantum h ([Wikipedia: Planck constant](#)); the energy continuum of the classical limit is therefore a coarse-grained description of a discrete substrate.

Principle 1.4 (Interactions). At low and intermediate energies the experimentally accessible interactions are: electromagnetism, gravitation, and at least one short-range nuclear interaction ([Wikipedia:](#)

Fundamental interaction). Of these, gauge invariance holds rigorously for electromagnetism ([Wikipedia: Maxwell's equations](#)) and approximately for the short-range channels, whereas the gravitational potential is not gauge-invariant in its Newtonian form ([Wikipedia: Newton's law of universal gravitation](#)).

1.1.2 The empirical particle spectrum

We assume the experimental phenomenology of elementary particles as established (Particle Data Group reviews; see also [Wikipedia: Standard Model](#), [Wikipedia: Neutrino](#)).

1.1.3 The unifying definition: material field quantum

The composite construction of this monograph requires a single concept that subsumes both massive (ponderable) particles and massless (imponderable) quanta of the interaction fields.

Definition 1.5 (Material field quantum). A *material field quantum* (denoted $\mathcal{M}_q$) is any irreducible quantum of energy carrying definite values of the conserved quantum numbers of Principle 1.1.

A quantum is *ponderable* iff its rest energy is non-zero, $m_0 > 0$; it is *imponderable* iff $m_0 = 0$. By the mass–energy equivalence $E = mc^2$ ¹ every $\mathcal{M}_q$ carries an inertial mass equivalent to its energy content, including the imponderable cases.

¹ [Wikipedia: Mass–energy equivalence](#)

Remark 1.6 (Terminology). The traditional designation “massless” is reserved here for the rest mass only; the inertial mass equivalent of the field energy is implicitly understood to be present.

1.2 Gravitomagnetic Extension of Newton's Law

1.2.1 Setup

Let a closed mass distribution $M_{(0)} = \text{const}$ of finite extent be enclosed in a Eulerian control volume V_0 in cartesian right-oriented coordinates $\mathbf{x} = (x_1, x_2, x_3) \in R_3$. Denote by $\mathbf{v} = \dot{\mathbf{x}}$ the velocity field of the mass-bearing discontinuities. The field of the configuration consists of an internal contribution μ_i acting in V_0 and an external contribution μ_e acting in $V \setminus V_0$. We posit, in the spirit of Definition 1.5, that the gravitational *field energy* itself contributes to the source.

Definition 1.7 (Effective gravitating mass). Let $\mu(\mathbf{x}, t) = \mu_i + \mu_e$ be the field-energy mass density and $M(\mathbf{x}, t) = \mu(\mathbf{x}, t) + M_{(0)}$ the effective gravitating mass. The associated densities are

$$\sigma_\mu = \partial\mu/\partial V, \quad \sigma = \partial M/\partial V, \quad \sigma_{(0)} = \partial M_{(0)}/\partial V. \quad (1.1)$$

1.2.2 The heuristic field equations

We propose the gravitomagnetic ansatz²

² The proportionality constant $\alpha > 0$ differs from the constant of the Poisson form of Newton's law and is left as a phenomenological parameter to be fixed empirically.

$$\boxed{\nabla \times \boldsymbol{\mu} \sim \alpha \mathbf{G} + \sigma \mathbf{v}, \quad \alpha \nabla \cdot \mathbf{G} = \sigma, \quad \alpha = \text{const} > 0.} \quad (1.2)$$

Proposition 1.1 (Continuity). *Mass conservation $d\sigma/dt = 0$ together with $\nabla \cdot \mathbf{v} = 0$ implies $\dot{\sigma} + \nabla \cdot (\sigma \mathbf{v}) = 0$. Taking ∂_t of (1.2)₂ then yields $\nabla \cdot (\alpha \dot{\mathbf{G}} + \sigma \mathbf{v}) = 0$, which is consistent with (1.2)₁ because $\nabla \cdot (\nabla \times \boldsymbol{\mu}) = 0$.*

1.2.3 Closing the system

Taking the curl of (1.2)₁ and using $\nabla \times (\nabla \times) = \nabla(\nabla \cdot) - \nabla^2$,

$$\alpha \nabla \times \dot{\mathbf{G}} + \nabla \times (\sigma \mathbf{v}) = b (\nabla(\nabla \cdot \boldsymbol{\mu}) - \nabla^2 \boldsymbol{\mu}), \quad (1.3)$$

with $b \neq 0$ an undetermined proportionality constant. We introduce the heuristic ansatz ³

$$\nabla^2 \boldsymbol{\mu} = a^2 \ddot{\boldsymbol{\mu}}, \quad 0 < |a^2| < \infty. \quad (1.4)$$

Setting $\beta = a^2/\alpha$ and using $\mathbf{w} = \nabla(\nabla \cdot \boldsymbol{\mu}) - b^{-1} \nabla \times (\sigma \mathbf{v})$, $\mathbf{w} = \dot{\sigma}_\mu \mathbf{f}(\mathbf{x})$ with $\mathbf{f} \cdot \nabla(\sigma - \sigma_{(0)}) = 0$, the system closes to

$$\boxed{\nabla \times \mathbf{G} \sim \beta \dot{\boldsymbol{\mu}} + (\sigma - \sigma_{(0)}) \mathbf{f}/\alpha, \quad \alpha \beta \nabla \cdot \dot{\boldsymbol{\mu}} = -(\sigma - \sigma_{(0)}) \nabla \cdot \mathbf{f}.} \quad (1.5)$$

Theorem 1.2 (Linearised propagation law). *In the weak-field regime $\nabla \sigma \rightarrow 0$, $\sigma - \sigma_{(0)} \rightarrow 0$, $\nabla \cdot \boldsymbol{\mu} \rightarrow 0$, $\mathbf{v} = 0$, $b = 1$, the system (1.2)–(1.5) reduces to*

$$\boxed{\nabla^2 \mathbf{p} + \alpha \beta \ddot{\mathbf{p}} = 0, \quad \mathbf{p} \equiv (\mathbf{G}, \boldsymbol{\mu}), \quad \omega^2 \alpha |\beta| = 1, \quad 0 < \omega < \infty.} \quad (1.6)$$

Remark 1.8 (Sign of β). For $\beta < 0$ equation (1.6) is the wave equation of a transverse gravitational radiation propagating with speed ω ; for $\beta > 0$ it is a four-dimensional Poisson-type equation describing a relaxed perturbation of a static potential. The empirical absence of detected transverse gravitational waves at the time of writing of the original manuscript favoured $\beta > 0$; the modern detection of gravitational waves ⁴ of course falsifies that assumption in its strong form. The derivation that follows keeps both signs open since the algebraic structure of the composite spacetime in Section 1.3 is insensitive to that choice.

1.3 Composite Spacetime and the Unified Lorentz Group

1.3.1 The two auxiliary spacetimes

Definition 1.9 (Electromagnetic auxiliary spacetime). The *electromagnetic* (Minkowski) spacetime is $R_{-4} = R_3 \times i c t$, with imaginary time

³ This step is genuinely heuristic: it postulates that $\boldsymbol{\mu}$ either supports radiative modes or, at worst, is a relaxed potential field. Both possibilities are kept open by leaving the sign of β free.

⁴ Wikipedia: Gravitational_wave; LIGO 2015.

coordinate $x_{-4} = i c t$. The propagation speed of electromagnetic radiation in vacuum $1/\sqrt{\mu_0 \epsilon_0} = c$ selects the imaginary unit in x_{-4} and the Lorentz group $\hat{A}_-$ acting on R_{-4} as

$$\hat{A}_- = \begin{pmatrix} \cos \psi_- & 0 & 0 & i \sin \psi_- \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i \sin \psi_- & 0 & 0 & \cos \psi_- \end{pmatrix}, \quad \tan \psi_- = i\beta_-, \quad c\beta_- = v. \quad (1.7)$$

Definition 1.10 (Gravitational auxiliary spacetime). The *gravitational* auxiliary spacetime is $R_{+4} = R_3 \times \omega t$, with real time coordinate $x_{+4} = \omega t$, where $\omega > 0$ is the (a priori unknown) propagation speed of (1.6). By construction, R_{+4} contains only gravitational field structures; no $\mathcal{M}_q$ resides in R_{+4} . The associated Lorentz-type group $\hat{A}_+$ acts as

$$\hat{A}_+ = \begin{pmatrix} \cos \psi_+ & 0 & 0 & \sin \psi_+ \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sin \psi_+ & 0 & 0 & \cos \psi_+ \end{pmatrix}, \quad \tan \psi_+ = \beta_+, \quad \omega\beta_+ = v. \quad (1.8)$$

Remark 1.11 (Auxiliary character). Neither R_{+4} nor R_{-4} is physically realised on its own: R_{+4} excludes ponderable matter, and a $\mathcal{M}_q$ -only spacetime without its gravitational field is equally unphysical. The two are auxiliary fibers that must be entangled.

1.3.2 The composite spacetime

Theorem 1.3 (Commutativity and orthogonality of the auxiliary groups). The matrices $\hat{A}_-, \hat{A}_+$ defined above satisfy

$$[\hat{A}_-, \hat{A}_+] = \hat{0}, \quad \hat{A}_- \hat{A}_-^T = \hat{E}, \quad \hat{A}_+ \hat{A}_+^T = \hat{E}. \quad (1.9)$$

Consequently the product $\hat{B} = \hat{A}_+ \hat{A}_-$ is again orthogonal ($\hat{B} \hat{B}^T = \hat{E}$) on the underlying complex field.

Sketch. Block multiplication: $\hat{A}_-$ acts non-trivially only on the (x_1, x_4) -block; $\hat{A}_+$ likewise on the same block but with real instead of imaginary rotation angle. Both blocks are special orthogonal on $\mathbb{C}^2$ and commute as $\text{SO}(2, \mathbb{C})$ elements with different rotation parameters. $\square$

Definition 1.12 (Physical four-spacetime). The physical four-spacetime is the composite

$$R_4 := R_{+4} \boxtimes R_{-4}, \quad (1.10)$$

defined as the quotient of $R_{+4} \times R_{-4}$ in which the two spatial R_3 blocks are identified, and the two time coordinates (x_{+4}, x_{-4}) are mapped to a single imaginary time $x_4 = i c t$ via $x_{\pm 4} \mapsto x_4$. The Lorentz transformation group on R_4 is $\hat{B}$, and the velocity bound for ponderable structures is the usual $0 \leq v < c$.

Remark 1.13 (“Entanglement”). The original term used in the source for the construction (1.10) is best rendered in modern terminology as a *fibred product*: the gravitational and electromagnetic sectors are not parallel copies, they share the spatial slice R_3 and project onto the same time coordinate.

1.3.3 The unified field tensor

Proposition 1.4 (Field tensors). On R_{-4} the electromagnetic field is described by an antihermitian two-tensor $F_{km} = -F_{mk}^*$, whose four-divergence gives the source current; on R_{+4} the gravitomagnetic field is an antisymmetric two-tensor $G_{km} = -G_{mk}$ obtained by assembling $(\mathbf{G}, \boldsymbol{\mu})$ in the standard antisymmetric way and whose four-divergence yields the extended Newton law of Section 1.2.

Definition 1.14 (Unified field tensor). The unified field tensor on R_4 is the non-Hermitian two-tensor

$$M_{km} \neq M_{mk}^* \quad (1.11)$$

that interpolates between F_{km} on the R_{-4} -fiber and G_{km} on the R_{+4} -fiber. Its iterated square produces the canonical energy-density tensor

$$T_{ik} = \sum_{m=1}^4 M_{im} M_{mk}, \quad T_{ik} \neq T_{ki}^*. \quad (1.12)$$

Remark 1.15 (Interpretation of T_{ik}). By Definition 1.5 and Principle 1.1, T_{ik} is the unified energy density of matter and field. Its non-Hermitian content captures the imaginary part associated with field energy, and is the source of the asymmetric (non-Riemannian) sector of the metric in Section 1.4.

1.4 Non-Hermitian Metric Structure

1.4.1 Channel decomposition of the metric

We collect, for each of the n_{int} interaction channels indexed by $1 \leq j \leq n_{\text{int}}$, a quadruple of geodesic coordinates $\xi_p^{(j)}$, $1 \leq p \leq 4$, oriented by unit cochains $\varepsilon_p^{(j)}$ (not necessarily orthonormal). Let m_{ngi} be the number of gauge-non-invariant channels and $n_{\text{int}} - m_{\text{ngi}}$ the number of gauge-invariant channels.

Definition 1.16 (Channel-resolved line element). Define the resultant differentials

$$dz_p^+ = \sum_{j=1}^{m_{\text{ngi}}} d\xi_p^{(j)}, \quad dz_p^- = \sum_{j=m_{\text{ngi}}+1}^{n_{\text{int}}} d\xi_p^{(j)}, \quad (1.13)$$

and the total line element $ds = \sum_{p=1}^4 (dz_p^+ + dz_p^-)$.

Theorem 1.5 (Three-block decomposition of the metric). The line element of R_4 decomposes as

$$ds^2 = (g_{ik}^{(1)} + g_{ik}^{(2)} + g_{ik}^{(3)}) dx^i dx^k \equiv g_{ik} dx^i dx^k, \quad (1.14)$$

with

$$g_{ik}^{(1)} = \sum_{p,q=1}^4 \epsilon_p^+ \cdot \epsilon_q^+ \partial_i z_p^+ \partial_k z_q^+, \quad g_{ik}^{(1)} = (g_{ki}^{(1)})^*, \quad (1.15)$$

$$g_{ik}^{(2)} = \sum_{p,q=1}^4 (\epsilon_p^+ \cdot \epsilon_q^- \partial_i z_p^+ \partial_k z_q^- + \epsilon_p^- \cdot \epsilon_q^+ \partial_i z_p^- \partial_k z_q^+), g_{ik}^{(2)} \neq (g_{ki}^{(2)})^*, \quad (1.16)$$

$$g_{ik}^{(3)} = \sum_{p,q=1}^4 \epsilon_p^- \cdot \epsilon_q^- \partial_i z_p^- \partial_k z_q^-, \quad g_{ik}^{(3)} = (g_{ki}^{(3)})^*. \quad (1.17)$$

The first and third blocks are Hermitian; the second is the non-Hermitian interference block.

Corollary 1.6 (Hermitian/antihermitian split). *Each tensor admits an additive decomposition*

$$g_{ik} = g_{ik}^+ + g_{ik}^-, \quad g_{ik}^+ = (g_{ki}^+)^*, \quad g_{ik}^- = -(g_{ki}^-)^*. \quad (1.18)$$

The contraction $g_{ik}^- dx^i dx^k = 0$ is identically zero, so the metric alone identifies only the Riemannian sector g_{ik}^+ .

1.4.2 Connection, curvature, and the field equation

Definition 1.17 (Non-Hermitian connection). Independently of the metric, the parallel-transport law on R_4 admits Christoffel-type cochains $\Gamma_{km}^i \neq (\Gamma_{mk}^i)^*$ with the additive split

$$\Gamma_{km}^i = \Gamma_{(+)km}^i + \Gamma_{(-)km}^i. \quad (1.19)$$

The associated Riemann tensor R_{kmp}^i , Ricci tensor $R_{km} = R_{kmi}^i$ and curvature scalar $R = g^{mk} R_{km}$ inherit non-Hermiticity and the same split.

Postulate 1.18 (Non-Hermitian Einstein equation). The interaction tensor and the curvature are equivalent (Heim's non-Hermitian extension of Einstein's equivalence principle):

$$\boxed{R_{ik} - \frac{1}{2} g_{ik} R \sim T_{ik}.} \quad (1.20)$$

In the Hermitian limit $g_{ik}^{(2)} = g_{ik}^{(3)} = 0$, $T_{ik} \rightarrow V_{ik} = V_{ki}^*$, this recovers the standard Einstein field equation $R_{ik}^{(1)} - \frac{1}{2} g_{ik}^{(1)} R^{(1)} \sim V_{ik}$ (Wikipedia: Einstein_field_equations).

Remark 1.19 (Status of (1.20)). Equation (1.20) is not divergence-free in the strong sense: neither the non-Hermitian Einstein tensor on the left nor T_{ik} on the right need vanish under $\nabla_i (\cdot)^{ik} = 0$. Principle 1.1 is therefore violated at the metric level; the violation is resolved only after the quantum-action principle of Section 1.5 is imposed, which will be the central technical claim of this chapter.

Remark 1.20 (Geodesic equation). With the connection of Definition 1.17, the geodesic equation retains its usual form,

$$\ddot{x}^i + \Gamma_{km}^i \dot{x}^k \dot{x}^m = 0, \quad (1.21)$$

on which the Hermitian part of the connection acts as a generalised “force” on test particles. This justifies reading g_{ik} as a generalised tensorial interaction potential, not merely as a metric.

1.5 Microscopic Discontinuities

1.5.1 From energies to actions

Take the trace of (1.20) on R_4 . Using $g_{k\cdot}^{\cdot k} = 4$ and $T = g^{ik} T_{ik} = T_{\cdot k}^{\cdot k}$,

$$R \sim -T, \quad (1.22)$$

whence (1.20) can be rewritten as

$$R_{ik} \sim T_{ik} - \frac{1}{2} g_{ik} T \equiv W_{ik}, \quad (1.23)$$

with W_{ik} an extended energy-density tensor whose components are themselves time-derivatives of an action 2-form ω_{ik} ⁵.

⁵ Because V is a 3-volume and the time coordinate enters via $x_4 = ic t$, the dimensional identification of W_{ik} with $ic (d\omega_{ik}/d\Omega)w$ is fixed up to the sign of the imaginary unit, which is the “orientation” of R_{-4} .

1.5.2 Quantisation of the action 2-form

By Principle 1.3 the components of every action 2-form are integer multiples of h , with the integers allowed to be complex with integer real and imaginary parts:

Postulate 1.21 (Action quantisation).

$$\omega_{ik} = h N_{ik}, \quad N_{ik} \in \mathbb{Z}[i]. \quad (1.24)$$

1.5.3 The discrete curvature–combinatorial relation

Definition 1.22 (Quantum-number density). Let $\Delta\Omega = w \Delta x^1 \Delta x^2 \Delta x^3 \Delta x^4$ be a finite 4-volume cell, with $w = \sqrt{-g}$, $g = |g_{ik}|_4 < 0$. The *quantum-number density* is

$$\eta_{ik} := \frac{\Delta N_{ik}}{\Delta\Omega}. \quad (1.25)$$

Theorem 1.7 (Discrete Einstein relation). Under Postulates 1.18–1.21, the non-Hermitian Einstein equation (1.20), expressed via its Ricci tensor and the energy reading (1.23), becomes the discrete relation

$$R_{ik} \sim w \eta_{ik}, \quad w = \sqrt{-g}, \quad \eta_{ik} = \frac{\Delta N_{ik}}{\Delta\Omega}, \quad \Delta\Omega = w \Delta x^1 \Delta x^2 \Delta x^3 \Delta x^4.$$

(1.26)

Sketch. Substituting $\omega_{ik} = h N_{ik}$ into $W_{ik} = ic (d\omega_{ik}/d\Omega)w$ and replacing the differential by the difference required by Principle 1.3 gives $W_{ik} = ic h w \eta_{ik}$. Insertion into (1.23) and dropping the universal constants into the proportionality yields (1.26). $\square$

Remark 1.23 (Two consequences). Equation (1.26) carries two distinct claims:

1. the Ricci tensor (and through it the metric and the connection) becomes a discrete, level-spaced quantity—the continuum metric is the coarse-grained limit of an integer-valued cochain;
2. the spacetime volume $\Delta\Omega$ that appears in (1.25) is itself a finite cell, suggesting that spacetime admits an irreducible elementary cell, the *metron*, which is the subject of Chapter 3.

Summary of notation introduced in this chapter

Symbol	Object
$\mathcal{M}_q$	Material field quantum (Definition 1.5)
$M_{(0)}$	Bare source mass
$\mu(\mathbf{x}, t)$	Gravitational field-energy mass density
$\mathbf{G}$	Gravitoelectric field vector
$\boldsymbol{\mu}$	Gravitomagnetic potential
α, β	Phenomenological constants in (1.2)–(1.5)
ω	Propagation speed of $\boldsymbol{\mu}$, (1.6)
R_{+4}	Gravitational auxiliary spacetime
R_{-4}	Electromagnetic Minkowski spacetime
$R_4 = R_{+4} \boxtimes R_{-4}$	Physical four-spacetime
$\hat{A}_+, \hat{A}_-, \hat{B}$	Lorentz-type matrices on R_{+4}, R_{-4}, R_4
F_{km}, G_{km}, M_{km}	EM, gravitomagnetic, unified field tensors
T_{ik}	Unified energy-density tensor
$g_{ik}^{(1,2,3)}, g_{ik}$	Metric three-block and non-Hermitian sum
$\Gamma^i_{(\pm)km}$	Hermitian/antihermitian Christoffel parts
$N_{ik} \in \mathbb{Z}[i]$	Gaussian-integer-valued action quanta
η_{ik}	Quantum-number density $\Delta N_{ik} / \Delta\Omega$
w	Volume factor $\sqrt{-g}$
$\Delta\Omega$	Discrete 4-cell volume

2

The World Tensor Manifold

Scope of the chapter. The discrete relation $R_{ik} \propto w \eta_{ik}$ obtained at the end of Chapter 1 carries two pieces of hidden geometric data. The first is a combinatorial *dimension selection rule*: counting how many independent non-zero eigenvalue spectra $\lambda_p(k, m)$ are compatible with the Hermiticity of the operators C_p that lift the connection to the microscopic regime forces the physical four-spacetime to be a Lorentz submanifold of a six-dimensional ambient manifold of signature $(+++--)$. The second is the existence of an irreducible elementary area $\tau > 0$, the *metron*, that quantises the geometry.

Four sections develop these consequences. Section 2.1 proves the dimension theorem $N = (n - 1)\sqrt{n}$ and selects $N = 6$ from $n = 4$. Section 2.2 constructs the world Lorentz group $\hat{C}(R_6)$, the world velocity $Y = v + iu$ and the anisotropy of x_4 . Section 2.3 establishes the Hermiticity of the world metric and lifts the unified field tensor to R_6 , leading to a three-partial-structure decomposition of the connection. Section 2.4 derives the modified gravitational potential $r\varphi(r) = \gamma M(r)(1 - r/\zeta)^2$, its outer extremum, the gravitational cutoff scale $\zeta \sim h^2/(\gamma m_0^3)$, and—through the empty-space limit of the lower realm boundary—the *metron formula* $\tau \omega c^2 = \pi \gamma \hbar$ together with the dimensional axiom $m \bmod p = 0$ that any metronised manifold must satisfy.

2.1 The Dimension Theorem

2.1.1 From the connection cochain to discrete eigenvalues

Let $\varphi_{km}^i \in C^\bullet(R_4)$ be the microscopic correspondent of the Christoffel cochain Γ_{km}^i (cf. Definition 1.17), normalised so that

$$J_{km}^i = \int_{\Omega} \varphi_{km}^i (\varphi_{km}^i)^* d\Omega < \infty, \quad J_{km}^i = 1. \quad (2.1)$$

The continuum limit $\varphi_{km}^i \rightarrow \Gamma_{km}^i$ is in general not convergent in the sense of (2.1); the convergent φ are therefore the microscopic objects.

Postulate 2.1 (Hermitian functional operators). There exist Hermitian functional operators C_p , $1 \leq p \leq n$, on the space of L^2 -cochains such that

$$\int_{\Omega} \left((\varphi_{km}^{(p)})^* C_{(p)} \varphi_{km}^{(p)} - \varphi_{km}^{(p)} (C_{(p)} \varphi_{km}^{(p)})^* \right) d\Omega = 0, \quad (2.2)$$

and the macroscopic limit reads $C_p \varphi_{km}^p \rightarrow R_{km}$, the Ricci tensor of (1.20).

Theorem 2.1 (Discrete eigenvalue equation). Equations (1.20), (1.26) and (2.2), together with the action quantisation (1.24), jointly force the

microscopic relation

$$\boxed{C_{(p)} \varphi_{km}^{(p)} = \lambda_{(p)}(k, m) \varphi_{km}^{(p)}, \quad \lambda_p(k, m) \in \mathbb{R}.} \quad (2.3)$$

Each spectrum $\{\lambda_p(k, m)\}$ is discrete and is the discrete structural level of the metric at the index triple (p, k, m) .

Sketch. Hermiticity of C_p forces real eigenvalues. The unitarity constraint $Q^i_{kmp} = C_{(p)} \varphi_{km}^{(p)} - \lambda_{(p)} \varphi_{km}^{(p)} = 0$ with $\text{tr}_{i=p} Q = 0$ is enforced by (1.24) (integer-valued action 2-forms cannot support non-trivial residual fluctuations). $\square$

2.1.2 Counting the empty spectra

The Riemann curvature has the antisymmetry $R^i_{kmp} = -R^i_{kpm}$ and the cyclic identity $R^i_{kmm} = 0$; these lift to the microscopic operators via the correspondence $C_p \varphi^i_{km} \rightarrow R^i_{kmp}$, yielding the *spectral vanishing rules*.

Proposition 2.2 (Vanishing rules). *For $1 \leq k, m, p \leq n$,*

$$\lambda_m(k, m) = 0 \quad (\text{from } R^i_{kmm} = 0), \quad (2.4)$$

$$\lambda_m(m, k) = 0 \quad (\text{from Hermiticity}), \quad (2.5)$$

$$\lambda_p(m, m) = 0, \quad p \neq m \quad (\text{from antisymmetric trace}). \quad (2.6)$$

Theorem 2.3 (Spectrum counting). *In a manifold of dimension n , the C_p -eigenvalue equation (2.3) carries n^3 spectra. Of these, (2.4)–(2.5) force $2n^2 - n$ to vanish (the $-n$ correcting the double counting at $k = m$) and (2.6) a further $n^2 - n$, leaving*

$$z(n) = n^3 - (2n^2 - n) - (n^2 - n) = n^3 - 3n^2 + 2n = 6 \binom{n}{3}. \quad (2.7)$$

2.1.3 The geometric square-root condition

The non-zero spectra carry energy-density information by $\lambda_p(k, m) \varphi_{km}^p \sim \varepsilon_{km}^p$. Invariance under the global Poincaré group requires the ε_{km}^p to assemble into the square matrix of a rank-2 tensor in some ambient manifold R_N with $N \geq n$. The matrix dimension must satisfy $z(n) = N^2 - 2N$, i.e. $N(N - 2) = z(n)$.

Theorem 2.4 (Dimension theorem).

$$\boxed{N = 1 + \sqrt{1 + r}, \quad r = 6 \binom{n}{3}, \quad 0 \leq n \leq N, \quad N \bmod 1 = 0.} \quad (2.8)$$

The equivalent factorised form is

$$N = (n - 1)\sqrt{n}, \quad (2.9)$$

so R_n embeds as a submanifold of R_N exactly when n is a perfect square.

Corollary 2.5 (The world dimensions). *Among the relevant cases:*

- $n = 4$: $r = 24$, $\sqrt{1+r} = 5$, $N = 6$.
- $n = 6$: $r = 120$, $\sqrt{1+r} = 11$, $N = 12$.

No other small n admits a positive integral lift. The physical four-spacetime R_4 therefore embeds into a six-dimensional world manifold R_6 , which in turn embeds into a twelve-dimensional manifold R_{12} .

Remark 2.2 (Lift cycle). The lift cycle $R_4 \subset R_6 \subset R_{12}$ admits a formal contraction $R_{12} \rightarrow R'_6$ via Lie group isomorphism $SU(2) \cong SO(3)$ ([Wikipedia: Spin_group](#)). The diagram

$$R_6 \rightarrow R_{12} \rightarrow R'_6 \neq R_6 \quad (2.10)$$

records that the contracted six-manifold is not isomorphic to the original one.

2.1.4 Vanishing pattern of the R_6 tensor

A direct consequence of (2.6) is that the R_6 rank-2 tensor T_{ik} inherits a fixed sparsity pattern: of its 36 components, 12 vanish identically.

Theorem 2.6 (R_6 sparsity). *The unified energy-density tensor on R_6 admits the block structure*

$$\hat{T} = \begin{pmatrix} T_{11} & T_{12} & T_{13} & T_{14} & 0 & 0 \\ T_{21} & T_{22} & T_{23} & T_{24} & 0 & 0 \\ T_{31} & T_{32} & T_{33} & T_{34} & 0 & 0 \\ T_{41} & T_{42} & T_{43} & T_{44} & T_{45} & T_{46} \\ 0 & 0 & 0 & T_{54} & T_{55} & T_{56} \\ 0 & 0 & 0 & T_{64} & T_{65} & T_{66} \end{pmatrix}, \quad (2.11)$$

i.e. the components $T_{j5} = T_{j6} = T_{5j} = T_{6j} = 0$ for $j = 1, 2, 3$ vanish. The hidden coordinates x_5, x_6 couple to the physical R_3 only through the time row/column T_{4i}, T_{i4} .

Remark 2.3 (Open future). Theorem 2.6 formalises the observation of v. Weizsäcker ([Wikipedia: Carl_Friedrich_von_Weizsäcker](#)) that the microphysical future is necessarily open: the “trans”-block $\{T_{55}, T_{56}, T_{65}, T_{66}\}$ can influence R_3 -data only via x_4 , so latent (hidden-dimensional) states are accessible to the physical sector only through temporal projection.

2.2 Construction of the World Manifold

2.2.1 Signature selection

By the dimensional structure of Chapter 1 (real R_3 coordinates and imaginary time $x_4 = ict$), the R_6 signature is one of:

$$a : (+++--+), \quad b : (+++---), \quad c : (+++-+), \quad d : (+++---).$$

Functional-determinant continuity across the embedding $R_4 \hookrightarrow R_6$ requires the algebraic character of $\sqrt{-g}$ to extend; this eliminates b and c .

Postulate 2.4 (Empirical signature). Closed-orbit stability of macroscopic gravitational systems ([Wikipedia: Kepler_orbit](#)) requires at most three real spatial dimensions; the same constraint at the microscopic level ([Wikipedia: Hydrogen_atom_orbitals](#)) excludes $p > 3$ real spatial dimensions. Signature a has five real dimensions and is thereby ruled out empirically. Signature $d : (+++--)$ is the unique surviving choice.

Definition 2.5 (World coordinates). The cartesian world chart is

$$\boxed{x_1, x_2, x_3 \in \mathbb{R} \hat{=} R_3, \quad x_4 = i c t, \quad x_5 = i \varepsilon, \quad x_6 = i \eta,} \quad (2.12)$$

with $e_i \cdot e_k = \delta_{ik}$ and $x_k = e_k x_k$. The hidden coordinates x_5, x_6 are imaginary; their physical semantics is to be determined.

2.2.2 The world Lorentz group

Definition 2.6 (World Lorentz matrix). For a relative motion along x_1 at velocity v , the six-dimensional Lorentz matrix is

$$\hat{C}(R_6) = \begin{pmatrix} \cos \psi & 0 & 0 & \sin \psi & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -\sin \psi & 0 & 0 & \cos \psi & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \tan \psi = i\beta, \quad u\beta = v, \quad \psi = -\psi^*. \quad (2.13)$$

It is orthogonal on $\mathbb{C}^6$ with $\det \hat{C} = 1$, and reduces in the 4×4 block to the standard Lorentz matrix of Definition 1.9.

Definition 2.7 (World velocity). The world velocity of a world line $s(t) \in R_6$ is

$$Y = \dot{s} = v + i u, \quad u^2 = c^2 + \dot{\varepsilon}^2 + \dot{\eta}^2. \quad (2.14)$$

In the photon-rest-frame analogue $\dot{\varepsilon} = \dot{\eta} = 0$, $u = c$ and Y reduces to its R_4 data; otherwise the hidden-coordinate rates contribute additively.

Proposition 2.7 (Tangent law and time anisotropy). With $Y \cos \alpha_k = \dot{x}_k$,

$$\sum_{k=1}^6 \cos^2 \alpha_k = 1. \quad (2.15)$$

For $j \neq 4$ the angles can be made orthogonal, $2\alpha_j = \pi$, by appropriate choice of frame. For $k = 4$ one has $u\sqrt{1 - \beta^2} \cos \alpha_4 = c$, which can never be $\pi/2$. Hence $\cos \alpha_4 = \pm 1$ in the rest frame, and R_6 is morphologically anisotropic in x_4 .

Corollary 2.8 (Parallel worlds). The $\pm$ in $\cos \alpha_4 = \pm 1$ implies the simultaneous existence of an antiparallel time-counted slice R_4^- , and the x_5, x_6 -translations give rise to families of parallel three-spaces $R_3^\pm(v_p)$, $p = 5, 6$. The physical $R_3 = R_3^+(0)$ is one element of these families.

Proposition 2.9 (Velocity-angle complementarity). *The angles between the world-line tangent ds and the real / imaginary parts of $\mathbf{Y}$ satisfy*

$$\varphi(v) - \varphi(u) = \pi/2. \quad (2.16)$$

2.2.3 World submanifolds

The 6-dimensional world cartesian chart supports a stratification:

$$\#V_1 = 6, \#V_2 = 15, \#V_3 = 20, \#V_4 = 15, \#V_5 = 6. \quad (2.17)$$

Among the twenty V_3 , exactly one is the real physical R_3 , one is the imaginary trans-space spanned by (x_4, x_5, x_6) , and the remaining eighteen are complex. Among the fifteen V_4 , one is the physical Minkowski spacetime $R_4 = R_{-4}$.

2.3 Hermitian World Structures

2.3.1 Hermiticity

Theorem 2.10 (Hermitian world metric). *Let $\eta_k(x) \in R_6$ be a geodesic chart with orientation cochains ε_k , $|\varepsilon_k| = 1$, and Gram matrix $\hat{A} = (\varepsilon_i \cdot \varepsilon_k) = \hat{A}^T$. The world metric is*

$$ds^2 = g_{ik} dx^i dx^k, \quad g_{ik} = \partial_i \boldsymbol{\eta} (\partial_k \boldsymbol{\eta})^* = g_{ki}^*. \quad (2.18)$$

The R_6 metric is Hermitian; the non-Hermitian metric of R_4 arises as a non-trivial slice of (2.18).

2.3.2 World volume and three-trace structure

Proposition 2.11 (Three Ricci traces). *From $R^i_{kmp} = \Gamma^i_{kp,m} - \Gamma^i_{km,p} + \Gamma^i_{ms}\Gamma^s_{kp} - \Gamma^i_{ps}\Gamma^s_{km}$ the three index contractions yield*

$$R^p_{kmp} = R_{km}, \quad R^m_{kmp} = -R_{kp}, \quad R^k_{kmp} = B_{mp} = -B_{pm}, \quad B_{mm} = 0. \quad (2.19)$$

The first is the Ricci tensor; the third is a genuine antisymmetric two-tensor with no analogue in symmetric Riemannian geometry.

2.3.3 The world Einstein equation

Postulate 2.8 (Hermitian world equation). The Hermitian metric satisfies the divergence-free relation

$$\boxed{R_{ik} - \frac{1}{2}g_{ik} R = \alpha T_{ik}, \quad \alpha = \text{const.}} \quad (2.20)$$

Equivalently, via $\text{tr} \hat{g} = 6$ and $T = g^{ik} T_{ik}$,

$$R_{ik} = \alpha \left(T_{ik} - \frac{1}{4} g_{ik} T \right). \quad (2.21)$$

Remark 2.9 (Consistency with the empirical principles). Equation (2.20) is divergence-free by Hermiticity, satisfying Principle 1.1; it is the Euler equation of an action functional (Principle 1.2); the correspondence to the discrete spectrum of Theorem 2.1 respects Principle 1.3; and its R_4 slice reproduces (d₁) and (d₂) of Principle 1.4 via Definitions 1.9–1.10.

2.3.4 World eigenvalue operator

Definition 2.10 (World connection cochain). Let $\{\{^i_m\}\}(x) \in C^\bullet(R_6)$ be the microscopic correspondent of Γ_{km}^i on the world manifold. Assemble the full schema

$$\widehat{\{\}} = \left(\{\{^i_m\}\}\right)_6, \quad C = \sum_{p=1}^6 C_p, \quad \lambda = \sum_{p=1}^6 e_p \lambda_p.$$

Theorem 2.12 (World operator equation). *The world cochain satisfies the rank-3 tensorial eigenvalue equation*

$$\boxed{C \widehat{\{\}} = \lambda \times \widehat{\{\}},} \quad (2.22)$$

whose first trace $\text{tr}(C \widehat{\{\}})_{km} = \lambda \widehat{\{\}}$ recovers R_{ik} in the macroscopic limit, and whose second (antisymmetric) trace gives the world B-tensor.

Corollary 2.13 (Orthogonality of the eigenvector to the log of the volume element). *With $w = \sqrt{-g}$,*

$$\lambda \perp \text{tr}(\widehat{\{\}}) \longrightarrow \nabla_6 \ln w. \quad (2.23)$$

That is, the eigenvalue vector λ in the world manifold is always normal to the gradient of the log-volume element.

2.3.5 Unified field tensor

Theorem 2.14 (Field tensor decomposition). *The Hermitian energy density $T_{ik} = T_{ki}^*$ admits an iterated square-root factorisation*

$$T_{ik} = \sum_{m=1}^6 M_{im} M_{mk}, \quad M_{ik} = -M_{ki}^*, \quad M_{kk} = 0, \quad (2.24)$$

i.e. a 6×6 antisymmetric anti-Hermitian unified field tensor with $\binom{6}{2} = 15$ independent components. Empirically these 15 components are assembled from five spatial vector fields:

$$\begin{array}{lll} (\mathbf{E}, \mathbf{H}) & \text{electromagnetic,} & \subset R_{-4}, \\ (\mathbf{G}, \boldsymbol{\mu}) & \text{gravitomagnetic,} & \subset R_{+4}, \\ \mathbf{K} & \text{EM-gravity coupling,} & \subset R_4. \end{array} \quad (2.25)$$

Sketch. Use the Hermitian positive-semidefiniteness of T_{ik} to diagonalise via a unitary $\hat{S}$: $\hat{S} \hat{T} \hat{S}^* = \hat{\Lambda}$. Set $\hat{\lambda}_{ii} = \sqrt{\Lambda_i}$ and recover $\hat{M} = \hat{S}^* \hat{\Lambda} \hat{S}$. The anti-Hermiticity $M_{ik} = -M_{ki}^*$ follows from the field-tensor structure of the electromagnetic and gravitomagnetic sectors. The fifth vector $\mathbf{K}$ saturates the remaining three components of $\binom{6}{2} = 15$ and represents the coupling of the EM field to its own gravitational self-energy. $\square$

Remark 2.11 (Two physical consequences). **(a)** The divergence freedom $\overline{\text{div}}_6 M_{ik} = 0$ recovers, in its R_4 block, the conservation of a generalised complex four-current—a neutral current and an electric current together. Together they describe every spacetime structural change in terms of partial derivatives in the hidden coordinates. **(b)** The grouping (2.25) into three sectors motivates the heuristic decomposition of the world metric into three partial structures.

2.3.6 Three-partial-structure decomposition

Postulate 2.12 (Three partial structures). There exist three partial connection cochains $\{\widehat{\cdot}_m^i\}(\mu)$, $\mu = 1, 2, 3$, with $\{\widehat{\cdot}_m^i\}(\mu) \neq \{\widehat{\cdot}_m^i\}(\mu)^*$ (non-Hermitian each), whose Hermitian sum is the world connection:

$$\widehat{\cdot} = \sum_{\mu=1}^3 \widehat{\cdot}_{(\mu)}^+, \quad \sum_{\mu=1}^3 \widehat{\cdot}_{(\mu)}^- = \hat{0}. \quad (2.26)$$

The three partial structures correspond physically to the three sectors of (2.25).

Remark 2.13 (Connection to Chapter 3). Equation (2.26) is the source-side specialisation of the polymetric composition law (3.45) for the case $\omega = 3$. The three partial vielbeins ${}^2\bar{\kappa}_{(1)}, {}^2\bar{\kappa}_{(2)}, {}^2\bar{\kappa}_{(3)}$ of Section 3.5 are identified here with the electromagnetic, gravitomagnetic and coupling sectors.

2.4 Gravitational Spatial Structures and the Metron

2.4.1 The corrected gravitational potential

Combining the gravitomagnetic ansatz (1.2)–(1.4) of Section 1.2 with the action quantisation (1.24) and the atomistic structure $M(r) = L m(r)$, $1 \leq L < \infty$, m the elementary nucleus mass, the spherically symmetric static potential acquires a finite-range correction.

Theorem 2.15 (Heim potential). *For a closed spherical source of total mass $M_0 = L m_0$ with gravitational range $\zeta < \infty$ set by the Compton wavelength of m_0 , the gravitational scalar field is*

$$\boxed{r \varphi(r) = \gamma M(r) (1 - r/\zeta)^2, \quad r q e^{-q} = A (1 - r/\zeta)^2,} \quad (2.27)$$

with $q = 1 - \sqrt{1 - \varepsilon \varphi}$, $8 c^2 \varepsilon = 3$, and the integration constant $16 c^2 A = 3 \gamma M_0$. The corresponding field strength is

$$r^2 G_g(r) = -\gamma M_0 (1 - (r/\zeta)^2) e^q (1 + q)^{-1}. \quad (2.28)$$

Remark 2.14 (Behaviour). For $r \ll \zeta$, $G_g \rightarrow -\gamma M_0/r^2$ (attractive Newton). At $r = \zeta$, $\varphi(\zeta) = 0$ and $G_g(\zeta) = 0$. For $r > \zeta$, $G_g > 0$ becomes repulsive ("latent"), saturating at the upper real-extremum R_+ of (2.27).

2.4.2 Derivation of ζ

Theorem 2.16 (Gravitational cutoff). *The gravitational range is set by*

$$\boxed{\gamma m_0^3 \zeta = h^2 Y_1, \quad 2 Y_1 m_0 c \zeta \gg h,} \quad (2.29)$$

where $Y_1 = 1$ for sufficiently weak fields and $Y_1 > 1$ near relativistically degenerate matter (neutron stars). For $m_0 = A_t m_N$, the mean nuclide mass, the numerical estimate is

$$A_t^3 \zeta \approx 46 \text{ Mpc}. \quad (2.30)$$

Remark 2.15 (Galaxy cluster prediction). Inserting the Russell composition of intra-galactic matter (70% H, 29% He, 1% heavies) gives a mean $A_t \in [1.9, 2.4]$, so $\zeta \in [10^7, 2 \times 10^7]$ ly for spiral nebulae. Beyond this scale, gravitational attraction ceases and the cosmic web becomes structurally chaotic—a prediction qualitatively consistent with the observed lack of bound super-clusters at scales above ~ 10 Mpc in modern surveys ([Wikipedia: Galaxy_filament](#)).

2.4.3 Upper and lower realm boundaries

Proposition 2.17 (Extrema of (2.27)). *The reality condition $1 - \varepsilon\varphi \geq 0$ on (2.27) cuts r at the two roots*

$$R_{\pm} = \zeta(1 + \alpha) \left(1 \pm \sqrt{1 - (1 + \alpha)^{-2}} \right), \quad \alpha e \gamma M_0 = 2 \omega c \zeta. \quad (2.31)$$

For $\alpha \gg 1$, $R_+ = 2\alpha\zeta$ and $R_- = \zeta/(2\alpha)$, so $R_+R_- = \zeta^2$ and

$$4 \omega c R_- = e \gamma M_0, \quad R_- \approx 3.78 \times 10^{-28} M_0 [\text{m/kg}], \quad (2.32)$$

which numerically agrees with the Schwarzschild radius up to a factor of order unity ([Wikipedia: Schwarzschild_radius](#)).

2.4.4 The metron formula

Theorem 2.18 (Metron formula). *The empty-space limit $\lambda \rightarrow \infty$ of the product λR_- along the spectral surface $\lambda R_- = \frac{\gamma h}{2 \omega c^2} (1 - R_-/\zeta_M)^2$ is finite and yields the elementary area*

$$\tau = \lim_{\lambda \rightarrow \infty} (\lambda R_-) = \frac{\gamma h}{2 \omega c^2}, \quad \tau \omega c^2 = \pi \gamma \hbar. \quad (2.33)$$

Numerically, $\tau \approx 6.15 \times 10^{-70} \text{ m}^2$.

Remark 2.16 (Status). The metron is the elementary area of the cellular complex on which the discrete-geometric machinery of Chapter 3 is built. It is not an “aether”; it is the irreducible cellular quantum of the world manifold.

2.4.5 The dimensional axiom for metronisability

Theorem 2.19 (Dimensional axiom). *A manifold R_m of dimension m admits a complete metronisation by p -dimensional metrons iff $m/p \in \mathbb{Z}_{\geq 1}$, i.e.*

$$M = m/p \geq 1, \quad M \bmod 1 = 0. \quad (2.34)$$

Sketch. The cellular volume element is $\Delta V_m \sim \tau^{m/p}$ (a single metron contributes τ to its own slice, and the cell counts assemble multiplicatively). For ΔV_m to be generated by an integer number of p -cells, the exponent m/p must itself be integral. $\square$

Corollary 2.20 (Six-dimensional and four-dimensional metronisability). *With $p = 2$: $M = 3$ in R_6 admits three partial metric structures (consistent with Postulate 2.12); $M = 2$ in R_4 admits two partial metric structures.*

Summary of notation introduced in this chapter

Symbol	Object
R_6, R_{12}	World manifold and its 12-dimensional lift
$x_5 = i\varepsilon$	Fifth (imaginary) world coordinate
$x_6 = i\eta$	Sixth (imaginary) world coordinate
$\hat{C}(R_6)$	Six-dimensional Lorentz matrix
$\mathbf{Y} = \mathbf{v} + i\mathbf{u}$	World velocity, $u^2 = c^2 + \varepsilon^2 + \eta^2$
C_p	Hermitian functional operator on cochains
φ_{km}^i	Convergent micro-correspondent of Γ_{km}^i
$\lambda_p(k, m)$	Real discrete eigenvalue spectrum of C_p
B_{mp}	Antisymmetric third Ricci trace (no GR analogue)
$\{\overset{i}{k}_m\}$	Microscopic connection cochain on R_6
$\{\}$	Full connection schema on R_6
$M_{ik} = -M_{ki}^*$	Unified anti-Hermitian field tensor on R_6
ζ	Gravitational cutoff range, $\gamma m_0^3 \zeta = h^2$
$R_{\pm}$	Outer / inner realm boundary of (2.27)
τ	Elementary cellular area; $\tau \omega c^2 = \pi \gamma \hbar$
$M = m/p$	Metronisability ratio of an R_m
ω	Propagation speed of μ ; $3\omega = 4c$

3

Discrete Structure Tensors

Scope of the chapter. This chapter develops the geometric substrate on which the remainder of the book stands: a cellular calculus in which space is composed of elementary area quanta and every field, operator, and differential identity is reformulated on this granular substrate. We work exclusively in the language of *discrete differential geometry*¹: a finite cell complex K carries cochains, a coboundary operator δ with $\delta^2 = 0$, a discrete Hodge star, a vielbein (frame field), and a connection cochain whose curvature is a discrete Riemann tensor. The construction is purely combinatorial; the smooth manifold appears only as the limit obtained when the cell scale shrinks to zero.

The chapter is organised in five sections. Section 3.1 introduces the cellular substrate and the discrete exterior calculus. Section 3.2 casts the analytic operators of the theory as cochain endomorphisms (*selectors*). Section 3.3 adds a discrete metric via a vielbein. Section 3.4 refines the construction to hierarchical (*fine* and *hyper*) cellular structures and gives the explicit metrisation procedure that translates any classical field equation into a cochain equation. Section 3.5 treats stacks of vielbeins (*polymetric composition*), defines the discrete connection (*fundamental condensor*) and its curvature (*structure compressor*), and closes with the composition law that builds a multi-frame connection from its partial components.

3.1 The Metron Complex and Discrete Exterior Calculus

3.1.1 Cells, metrons, and the dimensional selection rule

Let K be a finite regular CW complex of dimension N , with K_p its set of p -cells. We posit the existence of an elementary area quantum $\tau > 0$ and identify a p -cell $\sigma \in K_p$ with a *metron* of dimension p .²

Definition 3.1 (Metron). A *metron* is a p -cell $\sigma_p \in K_p$, $1 \leq p \leq N$, endowed with the invariant scalar $\tau > 0$ representing its p -dimensional content. Adjacent metrons share a $(p-1)$ -face; their common boundary is a geodesic of the discrete metric.

Definition 3.2 (Discrete manifold dimensions). A discrete manifold supporting p -dimensional metrons has parameters (N, p, M, L) subject to

$$N = pM, \quad L = \binom{N}{p}, \quad (3.1)$$

where $M \in \mathbb{Z}_{\geq 1}$ counts the layers of p -strata and L counts the independent p -dimensional coordinate planes.

The selection rule (3.1) is a strict combinatorial constraint: it counts the p -cells of a regular N -cube and forces N to be a multiple of p . It

¹The coboundary operator δ with $\delta^2 = 0$, a discrete Hodge star, a vielbein (frame field), and a connection cochain whose curvature is a discrete Riemann tensor. The construction is purely combinatorial; the smooth manifold appears only as the limit obtained when the cell scale shrinks to zero.

²The metron is the carrier of discrete area in this calculus; in the continuum limit $\tau \rightarrow 0$ it collapses to a point and ordinary differential geometry is recovered.

is the discrete analogue of the requirement that the volume form of an N -manifold be a non-degenerate top form built from p -blocks.

3.1.2 The discrete exterior derivative

A k -cochain is a real-valued function on K_k . The vector space of k -cochains is denoted $C^k(K)$. The basic differential operator is the backward finite difference along each coordinate direction.

Definition 3.3 (Discrete differential). Let $\varphi \in C^0(K)$ and let n index the cells of a one-dimensional sub-complex. The *discrete differential* is

$$\partial\varphi \equiv (d\varphi)(n) = \varphi(n) - \varphi(n-1). \quad (3.2)$$

On a multi-index complex with indices $1 \leq i \leq L$ we set $\partial_i\varphi = \varphi - \varphi(\dots, n_i - 1, \dots)$ and

$$\partial = \sum_{i=1}^L \partial_i, \quad [\partial_i, \partial_k] = 0. \quad (3.3)$$

Theorem 3.1 (Cohomology identity). *The discrete differential satisfies $d \circ d = 0$. Equivalently, on every 0-cochain φ and pair of directions (i, k) ,*

$$(\partial_i \times \partial_k)_- = 0. \quad (3.4)$$

Proof. Direct expansion of the backward differences: $(\partial_i \partial_k - \partial_k \partial_i)\varphi = [\varphi(n) - \varphi(n - e_i) - \varphi(n - e_k) + \varphi(n - e_i - e_k)] - [\varphi(n) - \varphi(n - e_k) - \varphi(n - e_i) + \varphi(n - e_k - e_i)] = 0$, where e_i is the unit shift along direction i . $\square$

3.1.3 The discrete primitive and Stokes pair

Definition 3.4 (Discrete primitive). Let $S \equiv \Sigma$ act on a k -cochain φ between two cell indices $n_1 \leq n_2$ by

$$S_{n_1}^{n_2} \varphi \partial n = \sum_{n=n_1}^{n_2} \varphi(n). \quad (3.5)$$

If $\varphi = \partial\phi$, then

$$J(n_1, n_2) = S_{n_1}^{n_2} \partial\phi \partial n = \phi(n_2) - \phi(n_1 - 1). \quad (3.6)$$

Proposition 3.2 (Stokes algebra). *For any cell indices $n_1 < v < n_2$,*

$$S_{n_1}^v + S_{v+1}^{n_2} = S_{n_1}^{n_2}, \quad S_{n_1+1}^{n_1} = 0, \quad S_{n_1}^{n_1} \varphi \partial n = \varphi(n_1), \quad (3.7)$$

and the integrability condition is $\partial\phi = \varphi$.

Theorem 3.3 (Iterated differential). *For every $k \in \mathbb{Z}_{\geq 0}$,*

$$\partial^k \varphi(n) = \sum_{\nu=0}^k (-1)^\nu \binom{k}{\nu} \varphi(n - \nu), \quad (3.8)$$

i.e. the k th finite difference with the k th row of Pascal's triangle as coefficients.

3.1.4 Discrete Leibniz

Proposition 3.4 (Product and quotient rules). *Let $C \in \mathbb{R}$ be a constant cochain and let $u, v \in C^0(K)$. Writing $Tv(n) = v(n-1)$,*

$$\partial \sum_j u_j = \sum_j \partial u_j, \quad \partial C = 0, \quad \partial(Cu) = C \partial u, \quad (3.9)$$

$$\partial(uv) = u \partial v + v \partial u - \partial u \partial v, \quad (3.10)$$

$$\partial\left(\frac{u}{v}\right) = \frac{1}{v} \begin{vmatrix} \partial u & \partial v \\ u & v \end{vmatrix} \begin{vmatrix} v & \partial v \\ 1 & 1 \end{vmatrix}^{-1}. \quad (3.11)$$

The bilinear correction $-\partial u \partial v$ in (3.10) is the obstruction to interpreting ∂ as a derivation of an associative algebra; it vanishes in the continuum limit.

Remark 3.5 (Discrete Morse data). A 0-cochain attains an interior extremum at n_{ext} when $\partial\varphi(n_{\text{ext}}) = 0$; the sign of $\partial^2\varphi$ at that point distinguishes minima ($\partial^2\varphi > 0$), maxima ($\partial^2\varphi < 0$) and inflection points ($\partial^2\varphi = 0$).

3.1.5 Semiclassical regime

When the cell scale is small relative to the field gradients, the discrete differential becomes a near-derivation.

Proposition 3.5 (Semiclassical lift). *Set $\partial_\epsilon = \alpha \partial$ with $\alpha = \sqrt[3]{\tau}$ and $0 < |\partial_\epsilon| \ll 1$. Then to leading order*

$$\partial_\epsilon \varphi \approx \varphi \partial_\epsilon \ln \varphi, \quad \partial_\epsilon e^\varphi \approx e^\varphi \partial_\epsilon \varphi. \quad (3.12)$$

3.1.6 Multi-cochain integration

For an L -fold index $\mathbf{n} = (n_1, \dots, n_L)$ in $\kappa_i \leq n_i \leq N_i$, the discrete primitive extends to

$$\phi = S_{1+\kappa_1}^{N_1} \cdots S_{1+\kappa_L}^{N_L} \varphi(n_i) \prod_{k=1}^L \partial n_k, \quad \partial_i n_k = \delta_{ik}. \quad (3.13)$$

Theorem 3.6 (Chain rule). *If $\varphi = \varphi(\psi_1, \dots, \psi_\lambda)$ with $\psi_j = \psi_j(\mathbf{n})$, then*

$$\partial \varphi = \sum_{j=1}^{\lambda} \partial_{\psi_j} \varphi \partial \psi_j, \quad \partial_{\psi_j} \varphi = \frac{\varphi - \varphi(\dots, \psi_j - \partial \psi_j, \dots)}{\partial \psi_j}. \quad (3.14)$$

3.2 Cochain Operators (Selectors)

A linear endomorphism $C : C^\bullet(K) \rightarrow C^\bullet(K)$ that maps one cellular field into another is called a *selector*. The selector calculus is the algebraic backbone of the rest of the chapter.

3.2.1 The selector algebra

Definition 3.6 (Selector). A *selector* is a linear cochain endomorphism. Selectors compose under the action symbol “;” so that $(C; n) = f(n)$ produces a new cochain.

We single out the *zero selector* $0; n = 0$, the *unit selector* $E; n = 1$, the *coordinate selector* $E; () ; n = n$, and the *constant selector* $C = a () / ()$ acting by $C; n = a$. Selectors form an associative, non-commutative algebra: $(C_i \times C_k)_{\pm} \neq 0$ in general.

Definition 3.7 (Oriented coordinate selector and frame). For each index i , let $\bar{e}_i$ be a unit orientation cochain. The *oriented coordinate selector* is $\bar{Z}(i) = \bar{e}_i ()_i$. The Gram matrix of orientations

$$\hat{A} = (\bar{e}_i \cdot \bar{e}_k)_{1 \leq i, k \leq L} \quad (3.15)$$

is in general field-valued, $\hat{A} = \hat{A}(\mathbf{n})$.

Definition 3.8 (Tensor selector). A tensor selector of order $m \leq L$ is the formal product

$${}^m \bar{C} = [{}^m] \left[\prod_{k=1}^m C_{i_k} \right]_L, \quad {}^m \bar{T} = {}^m \bar{C}; n, \quad 0 \leq m \leq L. \quad (3.16)$$

Proposition 3.7 (Symmetric/antisymmetric splitting). Define ${}^m \bar{C}_{\pm(l-1, l)} = \frac{1}{2} ({}^m \bar{C} \pm {}^m \bar{C}^{T_{l-1, l}})$. Then, in components,

$$2 C_{\pm(l-1, l)} = \prod_{k < l-1} C_{i_k}; (C_{l-1} \times C_l)_{\pm}; \prod_{k > l} C_{i_k}, \quad (3.17)$$

so that symmetry (resp. antisymmetry) at indices $(l-1, l)$ exists iff the anticommutator (resp. commutator) of the corresponding factor selectors is non-zero.

Proposition 3.8 (Contraction). For an order- m tensor selector the trace $\text{sp}_{j=l} {}^m \bar{C} = {}^{m-2} \bar{C}$ reduces the order by two; on the symmetric part it yields a contraction, and on the antisymmetric part it vanishes:

$$\text{sp}({}^m \bar{C}_{+(i, k)}) = {}^{m-2} \bar{C}, \quad \text{sp}({}^m \bar{C}_{-(i, k)}) = {}^{m-2} \bar{0}. \quad (3.18)$$

3.2.2 The cellular operators

Definition 3.9 (Gradient, divergence, curl). With $\bar{\partial} = \sum_{i=1}^L \bar{e}_i \partial_i$,

$$\text{GRAD } \varphi = \bar{\partial} \varphi, \quad \text{DIV } {}^m \bar{C} = \text{sp } \bar{\partial}; {}^m \bar{C}, \quad \text{ROT } {}^m \bar{C} = \bar{\partial}; {}^m \bar{C} - (\bar{\partial}; {}^m \bar{C})^T. \quad (3.19)$$

Theorem 3.9 (Discrete vector-calculus identities). On any metron complex,

$\text{ROT} \circ \text{GRAD} = 0, \quad \text{DIV} \circ \text{ROT} = \text{DIV} \circ \text{GRAD} - \text{GRAD} \circ \text{DIV}, \quad \text{DIV} \circ \overline{\text{DIV}} \circ \text{ROT} = 0, \quad \text{Im ROT} = 0.$
--

(3.20)

Theorem 3.10 (Discrete Stokes). For a 0-cochain ϕ , a 1-cochain $\bar{\phi}$ and a 2-cochain $\hat{\phi}$,

$$S(\text{GRAD } \phi \partial \bar{N}) = \text{const}, \quad (3.21)$$

$$S_{\Omega(L)} \text{DIV } \bar{\phi} \partial V = S_{\Omega(L-1)} \bar{\phi} \partial \bar{V}, \quad (3.22)$$

$$S S \text{ROT } \hat{\phi} \partial^2 \bar{F} = S \bar{\phi} \partial \bar{N}, \quad (3.23)$$

with $\partial \bar{N} = \sum_i \partial_i \bar{Z}(i)$, $\partial V = \prod_k \partial_k Z(k)$, and $\partial^2 \bar{F} = [\partial_i Z(i) \partial_k Z(k)]_L$.

3.2.3 Selector equations

A general differential selector equation of order N and degree M ties together a finite number of selectors p_κ via the ∂^ν operators with $1 \leq \nu \leq N$. As an example, the *creation selector* of the Fibonacci shift is

Example 3.10 (Creation selector). The recurrence $a_n = a_{n-1} + a_{n-2}$ is encoded by the selector equation

$$\partial^2 - 3\partial + () = 0, \quad 2\zeta = 1 + \sqrt{5}, \quad (3.24)$$

whose growth ratio $\zeta = (1 + \sqrt{5})/2$ is the golden ratio.

3.3 Discrete Metric and Vielbein

We now equip K with a discrete metric. As in continuum geometry it is described by a non-degenerate symmetric (possibly Hermitian) bilinear form on the tangent cochains, decomposable through a vielbein.

3.3.1 The metric cochain

Definition 3.11 (Discrete metric). On a primitively structured discrete manifold of dimension $N = pM$, the metric is the symmetric/Hermitian cochain

$$(\partial s)^2 = g_{ik} \partial x^i \partial x^k, \quad g_{ik} = \gamma_{ik}; n, \quad (3.25)$$

with $\partial x^k = \alpha^k = \kappa^k \sqrt{\tau}$. The companion constraint

$$\kappa^i \kappa^k \gamma_{ik} - \alpha(p, \tau) \frac{()}{()} = 0, \quad \alpha(2, \tau) = 1, \quad (3.26)$$

selects the admissible metric selectors.

Theorem 3.11 (Vielbein factorisation). *There exists a vector selector $\bar{\gamma}$ (the vielbein) and a tensor selector ${}^2\bar{\kappa}$ (the lattice kernel) such that*

$${}^2\bar{\gamma} = \bar{\gamma} \times \bar{\gamma} = \text{sp}({}^2\bar{\kappa} \times {}^2\bar{\kappa}), \quad {}^2\bar{\gamma}_{\pm ik} = \frac{1}{2}(\gamma_i \times \gamma_k)_{\pm}. \quad (3.27)$$

For non-Hermitian metrics, ${}^2\bar{\kappa} \neq {}^2\bar{\kappa}^\dagger$ in general.

3.3.2 The discrete volume form

Proposition 3.12 (Volume cochain). *With $\gamma; n = |\bar{\gamma}|_N; n = w^2$ and $w = W; n$,*

$$\partial V = \kappa \tau^M w, \quad V = \kappa \tau^M S W; n \partial n, \quad \kappa = \prod_{k=1}^N \kappa^k. \quad (3.28)$$

On a geodesic frame the volume element is a multiple of τ^M ; in the flat reference ($\gamma = E$) it reduces to τ^M , and for $p = N$ to τ itself.

3.4 Hyperstructure and the Metrisation Procedure

A *primitively structured* discrete manifold treats each p -block as a single strand of metrons. A genuine N -dimensional manifold needs a hierarchical refinement: each block is itself a p -dimensional sub-array of metrons, and the cellular layout is governed by a geodesic net induced by the metric.

3.4.1 Fine structure and hyper-selectors

Definition 3.12 (Fine structure). A *fine structure* of K assigns to each block index n_i a p -dimensional sub-index $(k_1^{(i)}, \dots, k_p^{(i)})$ in the range $P_l^{(i)} \leq k_l^{(i)} \leq Q_l^{(i)}$:

$$n_i(k_l^{(i)})_1^p = c_i; n, \quad c_i = c_i(\kappa_l^{(i)})_1^p. \quad (3.29)$$

For a multi-field $\varphi(n_i)_1^L$ on a fine-structured complex, the action factorises as $\varphi(n_i)_1^L = \phi; n$ with $\phi = \phi(K_k)_1^G$ the *field selector*.

Definition 3.13 (Hyper-selector and lattice selector). Choose geodesic coordinates $\zeta^k = \zeta^k(n_i)_1^L$ adapted to the metric. The *hyper-selectors* are the operators X^k such that $\zeta^k = X^k; n$. The *lattice selectors* of the flat reference are

$$C_k = \kappa_k \sqrt[p]{\tau} ()_k, \quad x_k = C_k; n. \quad (3.30)$$

A hyperstructure is non-trivial iff $C_k \neq X_k$; affinely equivalent to flat iff $C_k = \sum_l a_l X_l$ with constant a_l .

3.4.2 Spin and area cochains

Theorem 3.13 (Discrete spin 2-form). For two independent geodesic directions $\bar{\zeta}_\alpha, \bar{\zeta}_\beta$ within a p -block,

$$\partial^2 \bar{F}_{\alpha\beta} = \partial \bar{\zeta}_\alpha \times \partial \bar{\zeta}_\beta = -\partial^2 \bar{F}_{\beta\alpha}. \quad (3.31)$$

The cellular integral defines the discrete area selector

$${}^2\bar{F}_{\alpha\beta} = \text{SS} \partial \bar{\zeta}_\alpha \times \partial \bar{\zeta}_\beta = \text{ROT}_N \bar{\varphi}_{\alpha\beta}, \quad (3.32)$$

so that the antisymmetric area matrix $\hat{F} = ({}^2\bar{F}_{\alpha\beta})_p = \text{ROT}_N \hat{\varphi}$ identifies with the curl of a vector field. Setting $\hat{F} = \hat{s}; n$ and $\text{ROT}_N \hat{\varphi} = \text{ROT}_N \hat{\varphi}; n$ gives the spin selector matrix

$$\hat{s} = \text{ROT}_N \hat{\varphi}, \quad (\hat{s}; n)_{n=1} = \hat{\tau}, \quad (3.33)$$

where $\hat{\tau}$ is the spin matrix attached to a single metron.

3.4.3 The metrisation procedure

Given a classical (continuum) N -manifold equipped with metric ${}^2\bar{g}(x^l)$ and the data (τ, p, M) , the discrete reformulation proceeds in five steps.

- (a) *Dimensional check.* Verify or extend N, M so that $N = pM$ holds.
- (b) *Flat reference.* Build the lattice selectors C_k of the empty reference $R_{N(0)}$ and the spin selectors $\hat{s}, \hat{\tau}$; each metron p -block is bounded by $p(p-1)$ oriented faces.
- (c) *Metric structure.* Determine the smooth metric ${}^2\bar{g}(x^L)$ and its geodesic chart $\zeta^l(x^k)$ by solving $\Gamma_{kl}^i = 0$; metrize $x^k = C^k$; n and obtain the hyper-selectors $X^l(C^k)$ and the fundamental metric selector ${}^2\bar{\gamma}(C^k)$.
- (d) *Fine structure (when needed).* Define block volume selectors

$$F_i; n = S_{v=1}^{n_i} \int_{\tau} \prod_{l=1}^p d\zeta_{(i)}^l \partial v, \quad \tau c_i((\cdot)_{(i)}^l)_1^p = F_i(C_k)_1^N. \quad (3.34)$$

- (e) *Cochain lift of any PDE.* Translate every smooth field equation by

$$\varphi(x^k)_1^N \rightarrow \phi; n, \quad \frac{\partial \varphi}{\partial x^k} \rightarrow \left(\frac{\partial_k \phi}{\partial C^k} \right); n, \quad d\varphi \rightarrow \left(\sum_{k=1}^N \partial_{(C^k)} \phi \right); n, \quad \int \varphi d\psi \rightarrow S\phi; n \partial\psi. \quad (3.35)$$

3.5 Polymetric Composition, Connection, and Curvature

The construction generalises to manifolds whose metric is composed of ω partial vielbeins. This is the natural setting for theories in which several discrete frames coexist, each with its own hyperstructure.

3.5.1 Polymetric stacks and sieve operators

Let ${}^2\bar{\kappa}_{(\nu)}$, $1 \leq \nu \leq \omega = N/p$, be the lattice kernels of the partial structures. The partial vielbein equations are

$$\bar{\psi}_{(\nu)} = S {}^2\bar{\kappa}_{(\nu)}; (\cdot) \partial \bar{x}, \quad {}^2\bar{\gamma}_{(\nu\nu)} = \text{sp}({}^2\bar{\kappa}_{(\nu)} \times {}^2\bar{\kappa}_{(\nu)}). \quad (3.36)$$

Definition 3.14 (Sieve operator). A *sieve operator* $S(\mu)$ annihilates the non-flat content of one partial vielbein:

$$S(\mu); {}^2\bar{\kappa}_{(\mu)}; n = {}^2\bar{E}, \quad S(\mu)_1^s; F(\mu)_1^s = F(E), \quad (3.37)$$

and the inverse sieve $(\mu)S; f(E) = f(\mu)$ restores it.

3.5.2 The correlator

Theorem 3.14 (Polymetric correlator). For $\mu, \nu \in \{1, \dots, \omega\}$ define

$${}^2\bar{\gamma}_{(\mu\nu)} = \text{sp}({}^2\bar{\kappa}_{(\mu)} \times {}^2\bar{\kappa}_{(\nu)}), \quad \hat{\gamma} = ({}^2\bar{\gamma}_{(\mu\nu)})_{\omega}. \quad (3.38)$$

Diagonal entries are the partial metrics; off-diagonal entries are correlation mediators of partial structures, in general non-Hermitian and absent from the smooth continuum theory.

3.5.3 Condensation density

Proposition 3.15 (Cellular Jacobian). *Projecting a hyperstructure onto the flat reference creates a condensation density $K_k \geq 1$ such that*

$$\partial_l N_k = K_k \delta_{kl}, \quad \partial \bar{N} = \bar{K} \partial \bar{n}, \quad \bar{N} = S \bar{K} \partial \bar{n}, \quad {}^2\bar{K} = {}^2\bar{\kappa}; n. \quad (3.39)$$

The rows (resp. columns) of ${}^2\bar{K}$ are the vectors of the lattice kernel; the lattice kernel itself becomes the measure of geometric condensation.

3.5.4 Fundamental condensor (discrete connection)

Definition 3.15 (Fundamental condensor). With basis signature (ab) (the underlying pair of partial kernels) and contra signature (cd) (the lift to mixed variance), the *fundamental condensor* is the cochain triple

$$\Gamma_{pkl}^{(ab)}(\tau) = [pkl]_{(ab)}; n, \quad [\hat{ab}] = [{}^3][pkl]_{(ab)}]_{N'} \quad (3.40)$$

and its mixed-variance lift

$$\gamma_{(cd)}^{ip} [pkl]_{(ab)} = \left[\begin{smallmatrix} i \\ kl \end{smallmatrix} \right]_{(ab)}^{(cd)}, \quad [{}^3] \left[\left[\begin{smallmatrix} i \\ kl \end{smallmatrix} \right]_{(ab)}^{(cd)} \right]_N = \left[\begin{smallmatrix} \hat{cd} \\ ab \end{smallmatrix} \right]. \quad (3.41)$$

On a geodesic frame this cochain vanishes, $[\hat{cd} / ab]_{(\xi)} = \hat{0}$; on any non-trivial reference it does not.

Proposition 3.16 (Trace identity). *In the Hermitian, single-frame case ${}^2\bar{\gamma} = \text{sp}({}^2\bar{\kappa} \times {}^2\bar{\kappa})$ with $\gamma; n = e^{2\varphi}; n$,*

$$\partial_l \varphi = \alpha_l \left[\begin{smallmatrix} k \\ lk \end{smallmatrix} \right]_{(\kappa)+}^{(\kappa)}, \quad \ln \sqrt{|g|} = \varphi; n. \quad (3.42)$$

3.5.5 Condensor field selector (discrete covariant derivative)

Definition 3.16 (Condensor field selector). The discrete covariant derivative is the family

$$\left(\begin{smallmatrix} \beta \pm \\ \alpha \end{smallmatrix} \right)_{(\pm)k}^{(s_1)(s_2)} = \frac{1}{\alpha_k} \partial_k + \sum_{\lambda=\mu+1}^m ()_{\sigma} \left[\begin{smallmatrix} i_{\lambda} \\ \sigma k \end{smallmatrix} \right]_{(\alpha(\lambda))(\varepsilon_{\lambda}(s_1))}^{(\beta(\lambda))(\pm)}; n - \sum_{\lambda=1}^{\mu} ()_{\sigma} \left[\begin{smallmatrix} i_{\lambda} \\ \sigma k \end{smallmatrix} \right]_{(\alpha(\lambda))(\varepsilon_{\lambda}(s_2))}^{(\beta(\lambda))(\pm)}; n, \quad (3.43)$$

acting on a mixed-variance tensor field of order $m \leq N-1$, with modality index $\varepsilon \in \{1, \dots, 6\}$ labelling the six admissible signatures $\{[], []^{\dagger}, [], [], []^{\dagger}, 0[]\}$.

Theorem 3.17 (Flat limit). *As ${}^2\bar{\gamma} \rightarrow {}^2\bar{E}$ the condensor field selector collapses to the divergence operator:*

$$\lim_{{}^2\bar{\gamma} \rightarrow {}^2\bar{E}} (\kappa)_{(\pm)}^{(s_1)(s_2)} = \widehat{\text{DIV}}_{(x)}. \quad (3.44)$$

3.5.6 Composition law for the connection

Theorem 3.18 (Composition law). *The full discrete connection decomposes into contributions of the partial structures and their pairwise couplings:*

$$[i_{kl}] = \sum_{\alpha=1}^{\omega^4} \left(\left[\begin{smallmatrix} i \\ kl \end{smallmatrix} \right]_{(ab)}^{(cd)} + Q_m^i(\alpha) \left[\begin{smallmatrix} m \\ kl \end{smallmatrix} \right]_{(ab)}^{(cd)} \right), \quad \alpha \equiv \begin{pmatrix} cd \\ ab \end{pmatrix}. \quad (3.45)$$

The mixed-variance coupling cochain $Q_m^i(\alpha)$ vanishes iff the partial structures are decoupled.

3.5.7 Discrete Riemann curvature (structure compressor)

Definition 3.17 (Structure compressor). The discrete Riemann curvature (structure compressor) is the four-tensor cochain

$$\boxed{\varsigma_{klm}^i = \frac{1}{\alpha_l} \partial_l [i_{km}] - \frac{1}{\alpha_m} \partial_m [i_{kl}] + [i_{ls}] ; () [s_{km}] - [i_{ms}] ; () [s_{kl}].} \quad (3.46)$$

Theorem 3.19 (Continuum limit of curvature). The structure compressor reduces to the classical Riemann tensor in the continuum limit:

$$\lim_{\tau \rightarrow 0} {}^4\bar{\varsigma}; n = {}^4\bar{R}. \quad (3.47)$$

On the flat reference, ${}^4\bar{\varsigma} = \hat{0}$; on any hyperstructured reference, ${}^4\bar{\varsigma} \neq \hat{0}$.

Theorem 3.20 (Factorisation of the curvature). The structure compressor admits the factorisation

$${}^4\bar{\varsigma} = K; [\hat{\cdot}], \quad K_p^{(m)} = \left| \begin{array}{cc} \delta_m^p & 1 \\ () & \delta_p^m() \end{array} \right| ()^{(1,6)}_{(-)p'}, \quad (3.48)$$

in which K is the order-raising selector built from the discrete exterior derivative and the binary Kronecker selector.

3.5.8 Closure

Combining the composition law and the curvature factorisation, the discrete curvature of a polymetric stack splits additively into the curvatures of the partial vielbeins, plus an irreducible quadratic mixing term:

$${}^4\bar{\varsigma} = \sum_{\alpha=1}^{\omega^4} {}^4\bar{\varsigma}_{(\alpha)} + \mathcal{Q}([\hat{\cdot}]), \quad (3.49)$$

where $\mathcal{Q}$ is bilinear in the fundamental condensor.

The cochain calculus developed in this chapter is the substrate on which the dynamical content of the theory is built. Subsequent chapters lift the field equations and the conservation laws onto this discrete background; the cellular spin selector $\hat{s} = \text{ROT}_N \hat{\phi}$ acquires its physical interpretation as the elementary unit of intrinsic angular momentum, and the structure compressor (3.46) provides the discrete analogue of the curvature scalar that drives the gravitational sector.

Symbol summary

Symbol	Object
τ	Elementary area quantum (metron content)
K, K_p	Regular CW complex and its p -cells
N, p, M, L	Dimension, metron order, layers, p -strata count
φ, ϕ	0-cochain (scalar field), field selector
∂, d	Discrete differential (backward finite difference)
S	Discrete primitive operator (chain sum)
$\bar{e}_i, \hat{A}$	Frame field and its Gram matrix
$C_k, X^{\bar{k}}$	Lattice selector (flat), hyper-selector (curved)
$g_{ik}, {}^2\bar{\gamma}$	Metric and metric selector
${}^2\bar{\kappa}$	Lattice kernel (vielbein cochain)
∂V	Cellular volume cochain $= \kappa \tau^M w$
$\hat{s}, \hat{\tau}$	Spin selector and metron spin
$\hat{\gamma}, {}^2\bar{\gamma}_{(\mu\nu)}$	Polymetric correlator and its entries
$S(\mu), (\mu)S$	Sieve operator and its inverse
$\bar{K}, K_k$	Condensation density (cellular Jacobian)
$[i_{kl}]_{(ab)}^{(cd)}$	Fundamental condensor (discrete connection)
$(\kappa)_{(\pm)k}^{(s_1)(s_2)}$	Condensor field selector (discrete covariant derivative)
$\zeta^i_{klm}, {}^4\bar{\zeta}$	Structure compressor (discrete Riemann curvature)
$Q^i_{\bar{m}}(\alpha)$	Coupling cochain between partial structures

4

The World as Hyperstructure

Scope of the chapter. This chapter is the synthesis of everything that came before. The combinatorial selection rules of Chapter 2 forced the world manifold to be six-dimensional with metron count $L_2 = \binom{6}{2} = 15$ and metric layers $M = N/p = 3$. The cellular discrete calculus of Chapter 3 provides the machinery to lift the continuum equations of Chapter 1 onto that hyperstructured manifold. The product is a single *world selector equation* $L \cdot \{\widehat{}\} = {}^4\bar{0}$ whose solutions classify into four *hermetry forms*—imaginary self-condensations, time condensations, complex space condensations, and total space-time condensations—that we identify with gravitons, photons, neutral massive quanta, and charged massive quanta respectively. From these, several genuinely predictive numerical results follow: a discrete mass spectrum $m(n, q) = m(n) \eta_q$ with $\eta_q^4 \sqrt{4q^4 + \pi^4} = \pi$, an elementary electric charge $\varepsilon_{\pm} = \pm 3\sqrt{\hbar/R_-}/\pi^2$ reproducing the empirical value to within 0.125 ppm, the fine structure constant to within 0.0015%, and a Hubble radius matching modern measurements.

Six sections develop the synthesis. Section 4.1 constructs the empty reference tensorium and derives the world selector. Section 4.2 classifies the solutions into four hermetry forms. Section 4.3 solves the equations for each form, obtaining the canonical mass and charge spectra. Section 4.4 exploits the world metron to derive a cosmogonic polynomial $w^7 - w = q$ whose solution determines the universe's diameter, mass, and Hubble radius. Section 4.5 traces the foundations of quantum mechanics back to the dimensional cycle $R_4 \subset R_6 \rightarrow R_{12} \rightarrow R'_6$.

4.1 Structural Condensation Stages

4.1.1 The empty reference tensorium

Setting $p = 2$ in the dimension rule of Chapter 2, the world manifold R_6 admits $M = N/p = 3$ partial structures and $L_2 = \binom{6}{2} = 15$ elementary metronic tensors. We single out as reference the empty hyperstructure in which hyper- and lattice selectors coincide and the eigenvalue spectrum vanishes,

$$\bar{\lambda} = \bar{0}, \quad C_k = X_k, \quad {}^2\bar{\gamma}; n = {}^2\bar{E}. \quad (4.1)$$

With unit volume element $\Delta\Omega = \tau^3$ and orthonormal frame $\bar{e}_i \cdot \bar{e}_k = \delta_{ik}$, the empty cellular chart is

$$\bar{x}_k = \bar{X}_k; n, \quad \bar{X}_k = \bar{e}_k \alpha_k ()_k, \quad \alpha_k = \sqrt{\tau} \ (k \leq 3), \quad \alpha_k = i\sqrt{\tau} \ (k > 3). \quad (4.2)$$

4.1.2 The metron spin matrix

Each metron carries the discrete rotor of Section 3.4,

$${}^2\bar{\tau} = \text{ROT}_6\bar{\phi}; 1, \quad \tau^2\bar{P} = \text{ROT}_6\bar{\phi}, \quad (4.3)$$

whose components in the empty reference take the explicit antihermitian form

$${}^2\bar{\tau} = \tau \begin{pmatrix} 0 & 1 & 1 & i & i & i \\ -1 & 0 & 1 & i & i & i \\ -1 & -1 & 0 & i & i & i \\ -i & -i & -i & 0 & -1 & -1 \\ -i & -i & -i & 1 & 0 & -1 \\ -i & -i & -i & 1 & 1 & 0 \end{pmatrix}, \quad \det {}^2\bar{\tau} = 0. \quad (4.4)$$

Proposition 4.1 (Spin-isotropic mixing). *Three spin-orientation patterns satisfy the empty-world isotropy condition: (a) exogenic Δ_+ with all spins pointing outwards, (b) endogenic Δ_- with all spins pointing inwards, (c) a checkerboard mixing $R_{6(0)}$ in which each Δ_+ is bounded by 30 cells Δ_- and vice versa. Only case (c) carries a structural potency: exchanging two adjacent cells immediately generates a non-trivial spin configuration. The physical empty reference is therefore*

$$\Delta_{\pm} \hat{=} \pm {}^2\bar{\tau}, \quad R_{6(\pm)} \hat{=} \Delta_{\pm}, \quad R_{6(0)} = R_{6(+)} \oplus R_{6(-)}. \quad (4.5)$$

4.1.3 The world selector equation

Combining the discrete operator equation of Section 2.3, the structure compressor of Theorem 3.20, and the discrete connection from the metric (Section 3.5), one obtains:

Theorem 4.2 (World selector). *There exists a Hermitian functional selector*

$$L = K - \bar{\lambda} \times () \quad \text{such that} \quad L \cdot \widehat{\{ \}} = {}^4\bar{0}. \quad (4.6)$$

K is the discrete-curvature-raising selector of (3.48), $\bar{\lambda}$ are the discrete metric condensation eigenvalues, and $\widehat{\{ \}}$ is the fundamental condensor cochain. The equation (4.6) selects, from all possible $\widehat{\{ \}}$, those that describe physical structures of the world.

Remark 4.1 (Status). Equation (4.6) unifies the polymetric framework of Chapter 3 with the dimensional selection rule of Chapter 2 via the quantum-action principle. It is the central equation of the entire construction; everything in this chapter follows from it.

4.1.4 Two physical traces

Proposition 4.3 (Trace identities). *The two non-trivial matrix traces of (4.6) are*

$$\text{sp } K; \widehat{\{ \}} = \bar{\lambda} \widehat{\{ \}} \quad (\text{macroscopic Einstein equation}), \quad (4.7)$$

$$\bar{\lambda} \perp ()_{(-)}; \varphi, \quad \varphi; n = \ln \sqrt{-g} \quad (\text{orthogonality to the log-volume}). \quad (4.8)$$

The first reduces, in the continuum and Riemannian limit, to the Einstein equation $R_{ik} - \frac{1}{2}g_{ik}R \sim T_{ik}$ on the world manifold.

4.2 Hermetry Forms

4.2.1 Semantic architecture

Definition 4.2 (Semantic units). The world manifold R_6 admits four *semantic units* of which its hermetry forms are built:

$$s_{(1)} \equiv R_3, \quad s_{(2)} \equiv x_4, \quad s_{(3)} \equiv x_5, \quad s_{(4)} \equiv x_6. \quad (4.9)$$

A subspace $V_k \subset R_6$ is *hermetric* if its metric deviates from the empty reference; otherwise it is *antihermetric*.

4.2.2 Counting the hermetry forms

Theorem 4.4 (Hermetry classification). Among the $\sum_{k=1}^6 \binom{6}{k} = 63$ a priori possible hermetric subspaces of R_6 :

- Restricting to the semantic-unit refinements reduces the count to $\sum_{k=1}^4 \binom{4}{k} = 15$.
- The divergence-free condition $\overline{\text{div}}_6^2 \bar{M} = \bar{0}$ forces $s_{(3)}$ and/or $s_{(4)}$ to participate in every viable hermetry, leaving 12 candidates.
- The empty- λ rule on transcoordinates rules out the single-transcoordinate classes H_1 and H_2 , leaving $Z_4 = 4$ forms in class H_3 .

The four surviving hermetry forms are:

$a :$	$H(s_{(3)}, s_{(4)})$	imaginary self-condensation,
$b :$	$H(s_{(2)}, s_{(3)}, s_{(4)})$	imaginary time condensation,
$c :$	$H(s_{(1)}, s_{(3)}, s_{(4)})$	complex space condensation,
$d :$	$H(s_{(1)}, s_{(2)}, s_{(3)}, s_{(4)})$	complex space-time condensation.

(4.10)

4.2.3 The fundamental hermetry equation

For $q \geq 2$ hermetric coordinates, substituting the symmetry-induced condensor proportionalities into the world selector (4.6) yields the master partial-difference equation

$$((a(k, l) - 1) \underline{\partial}_l - \sum_{m \neq l} \underline{\partial}_m) ; \varphi_{kl} + \varphi_{kl}^2 = \lambda(k, l) \varphi_{kl}, \quad (4.11)$$

where $\varphi_{kl} = \phi_{kl}$; n is a linear condensor aggregate. A metronic integration of (4.11) yields the implicit first integral

$(E - \psi_{kl})^{\Lambda_{kl}+1} \psi_{kl}^{\Lambda_{kl}-1} = 2^{-2\Lambda_{kl}} C_{kl} e^{-\lambda_{kl}\mu}, \quad \psi_{kl} = \frac{b_i^{(kl)}}{\lambda(k, l)} \lceil i_{kl} \rceil.$
--

(4.12)

4.2.4 Eigenvalue spectra

Theorem 4.5 (Imaginary spectra at the extremum). *At the extremum $\psi_{kl} = 1$, $(\lambda_{kl}\mu)_{\text{ext}} = \alpha + i\beta$, the right-hand side of (4.12) must vanish. This forces $\cos \beta = 0$ and $\sin \beta = 0$, giving four eigenvalue spectra:*

$$\boxed{\varepsilon_+^{(\pm)} = \pm \frac{\pi}{2}(2(\cdot)_+ + E), \quad \varepsilon_-^{(\pm)} = \pm \pi(\cdot)_-, \quad \beta_+^{(\pm)} + \beta_-^{(\mp)} = \pm \frac{\pi}{2}.} \quad (4.13)$$

Only the imaginary part of $\lambda_{kl}\mu$ produces eigenvalues; the real part is inert. Imaginary semantic units are therefore the ultimate source of all discrete metric structure.

4.2.5 Four validity regions

The fundamental hermetry equation (4.11) admits distinct approximations in four regions of metronic phase space:

(1) *Metronic region.* Low cell counts; $\lambda_{kl}\mu; n \approx \Lambda_{kl}$ retained exactly via (4.12).

(2) *Approximation region.* High cell counts, $\Lambda_{kl} \rightarrow 0$:

$$\psi_{kl} \approx (E + C_{kl} e^{-\lambda_{kl}\mu})^{-1}. \quad (4.14)$$

(3) *Infinitesimal-microscopic region.* Cells negligible relative to the structure scale, $\tau \rightarrow 0$:

$$\psi_{kl}(x_s) = (1 + C_{kl} e^{\pm i\lambda_{kl}y})^{-1}, \quad y^2 = \varsigma^2 - r^2, \quad -\varsigma^2 = \sum_{s=4}^q x_s^2, \quad r^2 = \sum_{i=1}^3 x_i^2. \quad (4.15)$$

(4) *Macroscopic region.* $r \gg \varsigma$, eigenvalue spectrum condenses to a continuum:

$$\psi \approx (1 + C e^{\pm \lambda r})^{-1}. \quad (4.16)$$

4.2.6 The critical-distance law

Theorem 4.6 (Quantum number relation). *In the infinitesimal-microscopic region, the eigenvalue equation yields*

$$\boxed{r = \frac{2n_r + 1}{2n_\varsigma + 1} \varsigma, \quad n_r < n_\varsigma, \quad n_r, n_\varsigma \in \mathbb{Z}_{\geq 0}.} \quad (4.17)$$

This is the discrete relation that quantises the spatial extent of every hermetric structure in terms of its hidden-coordinate radius ς .

4.3 Hermetric Elementary Structures

4.3.1 Form a: gravitons

Theorem 4.7 (Graviton interpretation). *For the imaginary self-condensation a , the symmetry $a^2 = b^2 = -1$, $a = b = -1$ together with the R_3 -distance $-\varsigma^2 = x_5^2 + x_6^2$ yields the universal level law*

$$H_{(1)} = \pi \varepsilon n_1, \quad H_{(2)} = \pi \varepsilon n_2, \quad \varsigma \geq \varepsilon = \text{const} > 0. \quad (4.18)$$

The world lines of a -structures are geodesic null lines of the R_4 -cone. When projected onto R_{+4} they appear as discrete quanta of a propagating gravitational field with speed $\omega = 4c/3$; these are the gravitons.

Remark 4.3 (Universal gravitation). The composition law

$$a \hat{=} G(x_5, x_6) \rightleftharpoons P(x_4, x_5, x_6) \hat{=} b \quad (4.19)$$

implies that every elementary quantum carries an attached graviton field; gravitational interaction cannot be screened.

4.3.2 Form b : photons

Theorem 4.8 (Photon interpretation). For the imaginary time condensation b with $\mu = i\sqrt{\tau} \sum_{k=4}^6 ()_k$, the projected wave equation reduces in the linearised limit to

$$\nabla^2 P - \frac{1}{c^2} \ddot{P} \sim P, \quad (4.20)$$

which is the standard electromagnetic wave equation ([Wikipedia: Maxwell's equations](#)); b -structures therefore describe photons.

4.3.3 Form c : neutral massive quanta

Theorem 4.9 (Mass spectrum of c -terms). For complex space condensations, the modality $K_{23} = \lambda_C / \Lambda = r_0 / \zeta$ is the unique survivor of the available 16 interpretive modalities. Combined with the metron formula $\gamma m_0^3 \zeta = h^2$, it yields the discrete mass spectrum

$$m(n) = \mu \frac{\sqrt[4]{2n}}{\sqrt{2n-1}}, \quad \mu = \sqrt{\frac{ch}{\gamma}}, \quad n \in \mathbb{Z}_{\geq 1}. \quad (4.21)$$

μ is the Planck mass. The maximum is $m_{\max} = m(1) = \mu \sqrt[4]{2}$, the upper limit of every possible energy quantum.

Corollary 4.10 (Interpretation). c -terms are neutral massive matter quanta (*neutrokorpuskeln*). The projection of x_6 onto R_3 is the half-Compton wavelength $\eta = \lambda/2 = h/(2mc)$; the projection of x_5 is the attractive-field cutoff $r = \varepsilon = \Lambda/2$ with $\Lambda = 2\zeta$.

4.3.4 Form d : charged massive quanta

The total R_6 -hermetry d couples a complex space condensation c to a time condensation b , with $-\zeta_d^2 = \varepsilon^2 + \eta^2 + c^2 t^2$ growing with t . The attached b -mode is the source of an electromagnetic field.

Theorem 4.11 (Mass correction by electric charge). The mass ratio between a d -term and its c -companion is

$$m(n, q) = m(n) \eta_q, \quad \eta_q \sqrt[4]{4q^4 + \pi^4} = \pi, \quad q \in \mathbb{Z}_{\geq 0}. \quad (4.22)$$

q is the electric charge quantum number; $q = 0$ recovers the neutral case.

4.3.5 The elementary electric charge

Theorem 4.12 (Quantised electric charge). *From the requirement that $\sin \varphi_{\max} = \pm 1$ for $U_{\max} = 1 + \sin^2 \varphi$, the minimum electric charge is*

$$\pi^2 \varepsilon_{\pm} = \pm 3 \sqrt{\hbar/R_-}, \quad R_- = \sqrt{\mu_0/\varepsilon_0} \text{ (vacuum wave impedance).} \quad (4.23)$$

A speculative tripartite decomposition gives $\varepsilon_{\pm} = 3 C_{\pm}$ with $\pi^2 C_{\pm} = \pm \sqrt{\hbar/R_-}$.

Numerically $\varepsilon_{\pm}$ matches the empirical electron charge to within +0.125 ppm.

4.3.6 The fine-structure constant

Theorem 4.13 (Theoretical α). *Combining (4.23) with the empirical relation $e_{\pm}^2 \sim \alpha$ via the convex-combination potential $2V_{ee} = V_{rr} + V_{ww}$ yields*

$$(2\pi)^5 \alpha = 9 \vartheta, \quad \vartheta = 5\eta + 2\sqrt{\eta} + 1, \quad \eta^4 \sqrt{4 + \pi^4} = \pi. \quad (4.24)$$

α is therefore a purely transcendental cosmological constant determined by π alone.

Numerically $1/\alpha = 137.038$, within 0.0015% of the modern measured value ([Wikipedia: Fine-structure_constant](#)); the sub-per-mille discrepancy is closed by the cellular correction derived in Volume II.

4.3.7 Pseudo-continuum and superposed discrete spectrum

Remark 4.4 (Two spectra in one). (4.21) is a dense “pseudo-continuum” containing every admissible field mass (including the imponderable gravitons and photons of forms a and b). The truly observable particle masses sit as a discrete sub-spectrum on top of it, extracted by the additional charge-quantisation (4.22). The separation of pseudo-continuum from true ponderable spectrum is the major task of Volume II.

4.4 Cosmogonic Consequences of the World Metron

4.4.1 Metron deformation by mass

Theorem 4.14 (Mass-deformed metron). *A non-zero source mass M_0 deforms the cellular volume from τ to*

$$F = \tau (1 - R_-/\zeta)^2 = \tau (1 - \alpha A)^2, \quad A = M^4, \quad \alpha \hat{=} \frac{8}{3e} \mu^{-4}. \quad (4.25)$$

The cellular length elements split symmetrically:

$$\delta s_{1,2} = \delta s_0 (1 \pm u), \quad u = (M_0/M)^4 = R_-/\zeta, \quad \delta s_1 + \delta s_2 = 2 \delta s_0. \quad (4.26)$$

Setting $a = l'/l = 1 - R_-/\zeta \in [0, 1]$, the upper bound $a = 0$ gives a maximum mass beyond which the elementary attractive field ceases to exist.

4.4.2 The cosmogonic polynomial

Theorem 4.15 (Universe diameter polynomial). *Combining the modified Newton potential of Theorem 2.15, the dimensional reduction $R_6 \rightarrow R_3$ via $\pi D^2 = n\tau$, and the substitution $2w^2 = f^6 \sqrt{6n/\pi}$ yields the cosmogonic Galois polynomial*

$$w^7 - w = q, \quad q = a^3 \sqrt{n}, \quad a\sqrt{\pi} = \sqrt[6]{\pi/6}. \quad (4.27)$$

Proposition 4.16 (Number of real roots). *At $n = 1$ ($t = 0$), (4.27) reduces to $\eta^7 - \eta = a$ and has three real roots $\eta_1 < \eta_2 < \eta_3$ plus two complex-conjugate pairs. For every $n > 1$ ($t > 0$) it has exactly one real root plus three complex pairs. The trinity of solutions at $t = 0$ persists as a three-step actualisation of a single root.*

4.4.3 The cosmogonic sphere trinity

Theorem 4.17 (Sphere trinity). *At $t = 0$, the three real roots produce the trinity of diameters*

$$\{D_f, D_m, D_p\} \quad \text{and} \quad \{d_f, d_m, d_p\}, \quad (4.28)$$

satisfying $D_j d_j = D_0^2 = \text{const.}$ Numerically $D_0^2 \approx 0.652 \text{ m}^2$ in the metron unit system, with $d_f = 0.17617186$, $d_m = 0.61267970$, $d_p = 0.71660232$ in metres.

4.4.4 The Hubble radius prediction

Theorem 4.18 (Hubble radius from cosmogenesis). *Setting $u_2 = \sqrt[4]{2} \cdot 2/\sqrt{3} - 1 \approx 0.373$ and $M_1(T) = 2u_2 \mu^4 / m_a^3$ with $m_a = m_N/3$ the mean quark mass, the cosmogenesis volume gives an optical radius*

$$2c^2 R = 3e\gamma M_1(T), \quad R \approx 1.15 \times 10^{26} \text{ m} \approx 13.4 \times 10^9 \text{ ly}, \quad (4.29)$$

with mean density $\sigma \approx 5.94 \times 10^{-27} \text{ kg m}^{-3}$ and Hubble constant $H \approx 56.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

These values agree with modern astrophysical measurements ([Wikipedia: Age_of_the_universe](#), [Wikipedia: Hubble%27s law](#)).

4.4.5 Subuniverse structure

Theorem 4.19 (Two-tier cosmic structure). *The quadratic relation $x_1 x_2 = 1$ in (4.27) splits the total universe into subuniverses of diameter $D' = 0.466 \times 10^{35} \text{ m}$. The total universe ($D \approx 6 \times 10^{125} \text{ m}$) contains $(D/D')^3 \approx 2.16 \times 10^{273}$ subuniverses; each subuniverse contains $(D'/d_0)^3 \approx 2.92 \times 10^{24}$ optical universes of diameter $d_0 = 2Y \approx 3.26 \times 10^{26} \text{ m}$.*

4.4.6 Electron mass from the secondary cosmogonic root

Theorem 4.20 (Electron mass). *With $m_{\min}^4 = \mu^4 \sqrt{\tau'/\tau}$ obtained from the secondary cosmogonic root, and applying the quantum-electrodynamic mass correction $\Delta = \alpha' m / (2\pi) + m\epsilon$ with $\epsilon = 0$,*

$$m_e = m_{\min} \left(1 + \frac{\alpha'}{2\pi}\right) \approx 9.10924 \times 10^{-31} \text{ kg}. \quad (4.30)$$

The measured value is 9.10939×10^{-31} kg; the discrepancy is 0.016 ppm.

4.4.7 Inflationary phase

Remark 4.5 (Inflation). The material cosmogonesis epoch $T - T_1 \sim 10^{18}$ s ($T/(T - T_1) \sim 10^{97}$) is the period in which the maximon $m_M = m(1,0) = \mu \sqrt[4]{2}$ decays into a cascade $m_M \rightarrow \nu m_0$ of intermediate-mass quanta. The phase $u_1 \rightarrow u_2 \approx 0.373$ corresponds to the modern theory's inflationary phase ([Wikipedia: Inflation_%28cosmology%29](#)), during which X-boson decays release vacuum energy into matter generation.

4.5 Origins of the Quantum Principle

4.5.1 The Galois symmetry of the cosmogonic polynomial

Theorem 4.21 (Galois group of (4.27)). The polynomial $f(w) = w^7 - w - q$ has Galois group S_7 . At $t = 0$ the symmetry decomposes as

$$G = S_3 \times (S_2 \times S_2) \quad (3 \text{ real, } 2+2 \text{ complex roots}). \quad (4.31)$$

For $t > \vartheta$ (after the first chronon),

$$H = S_1 \times (S_3 \times S_3) \quad (1 \text{ real, } 3+3 \text{ complex roots}). \quad (4.32)$$

The time-independent total symmetry is

$$\overline{H} \cong (S_1 \times S_1) \times (S_2 \times S_2) \times (S_3 \times S_3) \cong (S_3 \times S_1 \times S_2) \times (S_3 \times S_1 \times S_2). \quad (4.33)$$

Corollary 4.22 (Twelve-dimensional representation). The dimension of the representation space of $\overline{H}$ is exactly 12, recovering the upper dimensional level R_{12} of Corollary 2.5. The representation space of the related $S_3 \times S_1 \times S_2$ block has dimension 6, and the extended group $U \times U$ acts naturally on a 28-dimensional representation.

Remark 4.6 (Comparison with superstring). The reductions $R_n \rightarrow R_{n-2}$ map $(12, 28) \rightarrow (10, 26)$, the characteristic dimensions of the heterotic superstring theory ([Wikipedia: Heterotic_string_theory](#)). The recovery of these numbers from a purely combinatorial symmetry argument suggests a structural connection that deserves further investigation.

4.5.2 The twelve R_{12} coordinates

Theorem 4.23 (R_{12} chart). The cellular elements of R_{12} are

$$\delta x_k = \delta s_0 = \sqrt{\tau}, \quad k \leq 3, \quad (4.34)$$

$$\delta x_k = i \delta s_0, \quad 4 \leq k \leq 6, \quad (4.35)$$

$$\delta x_7 = i \delta l_0 \exp(2\pi m), \quad m \in \mathbb{Z}_{\leq 0}, \quad (4.36)$$

$$\delta x_8 = i \delta l_0 \exp(-2\pi m), \quad m \in \mathbb{Z}_{\geq 0}, \quad (4.37)$$

$$\delta x_{9,10} = i \delta l_\alpha, \quad \delta x_{11,12} = i \delta l_\beta, \quad (4.38)$$

with $\delta l_0 = \sqrt{\pi D_1 d_1} \approx 1.4312$ m the time-independent organisational length.

4.5.3 The grand-unification length

Theorem 4.24 (GUT length). *The characteristic length δl_v of the quantum theory is the $m = 12$ value of (4.37):*

$$\delta l_v = \delta l_0 e^{-24\pi} = a \delta s_0, \quad a \approx 103.785, \quad \delta l_v \approx 6.07 \times 10^{-17} \text{ m.} \quad (4.39)$$

This is approximately the grand-unification length of modern electroweak-strong theory (Wikipedia: [Grand_Unified_Theory](#)). A secondary value $28 \delta l_v = 2906 \delta s_0$ involves the dimensions 12 and 28 of $\overline{H}$.

4.5.4 Derivation of the uncertainty relation

Theorem 4.25 (Heisenberg uncertainty). *Combining the elementary length quantisation $\Delta s_8 = \pm i k \delta l_v$ with the metric corrections $\Delta r' \approx \Delta r(1 - E_G/E)$ and $\Delta t' \approx \Delta t(1 + E_G/E)$ from special and general relativity, one obtains the discrete invariance $\Delta \bar{E} \Delta \bar{t} \geq a/c$. Identifying $a = c\hbar$ via Principle 1.3, the universal uncertainty relation is*

$$\Delta \Gamma \Delta \Lambda \geq \hbar \quad \text{for any canonically conjugate } \Gamma, \Lambda. \quad (4.40)$$

The energy quantisation $E_n = nh\nu$ and de Broglie relation $p\lambda = h$ follow as corollaries.

Corollary 4.26 (Wave-particle duality). *The relation $\Delta s_8 = \pm \Delta s_4$ on geodesic null lines forces a probabilistic correspondence between the hidden (x_7, x_8) -actions and the physical time x_4 , recovering the Born rule and the Copenhagen interpretation (Wikipedia: [Copenhagen_interpretation](#)) from the geometry of $R_{12} \rightarrow R_6$.*

4.5.5 The source of quantisation

Remark 4.7 (Status of quantum mechanics). Theorem 4.25 shows that quantum mechanics is neither incomplete nor fundamental. It is the necessary projection of a $G_4(x_9, \dots, x_{12}) \rightarrow R_4$ chain in which the timeless coordinates of R_{12} act as the source of discrete actualisation in physical spacetime. Equivalently, the uncertainty relation is the residue, in R_4 , of the $R_6 \rightarrow R_{12} \rightarrow R'_6$ dimensional cycle of (2.10).

4.5.6 Gravitation and quantum mechanics cannot be unified in R_4

Theorem 4.27 (No unification in R_4 or R_6). *Gravitation is governed by $\delta s_0 = \sqrt{\tau}$ and quantum theory by $\delta l_v = a \delta s_0$. Their unification requires access to the timeless coordinates of $G_4(x_9, \dots, x_{12})$, which lie outside both R_4 and R_6 . No theory restricted to R_4 or R_6 can therefore achieve the unification.*

Summary of notation introduced in this chapter

Symbol	Object
$R_{6(0)} = R_{6(+)} \oplus R_{6(-)}$	Empty checkerboard reference tensorium
$2\bar{\tau}$	Metron spin matrix
$L = K - \bar{\lambda} \times ()$	World selector
$\widehat{\{ \}}$	Fundamental condensor cochain
$s_{(j)}$	Semantic architectural unit
$H_j = H(\dots)$	Hermetry on a semantic unit subset
a, b, c, d	Hermetry forms (graviton, photon, neutral $\mathcal{M}_q$, charged $\mathcal{M}_q$)
ψ_{kl}	Linear condensor aggregate
$\Lambda_{kl}, \lambda_{kl}$	Modified and reduced eigenvalue spectra
μ	Linear glitter-selector aggregate
$\delta s_{1,2} = \delta s_0(1 \pm u)$	Mass-deformed metronic length elements
$u = (M_0/M)^4$	Cellular deformation parameter
$m(n) = \mu \sqrt[4]{2n} / \sqrt{2n-1}$	Neutral mass spectrum (Planck mass μ)
η_q	Charge-correction factor
$\varepsilon_{\pm}$	Elementary electric charge
α	Fine-structure constant from (4.24)
$\delta l_0 = \sqrt{\pi D_1 d_1}$	Organisational length (≈ 1.43 m)
$\delta l_v = a \delta s_0, a \approx 103.8$	Grand-unification length
$w^7 - w = q$	Cosmogonic Galois polynomial
$\{D_f, D_m, D_p\}$	Cosmogonic sphere trinity
$R \approx 13.4 \text{ Gly}$	Predicted Hubble radius
$\overline{\overline{H}}$	R_{12} -representation group
$G_4(x_9, \dots, x_{12})$	Timeless coordinate sector

5

Cosmology

Scope of the chapter. The mass spectrum of Theorem 4.9 is in fact a *pseudo-continuum*: it contains, side by side, the imponderable a and b hermetries (gravitons and photons) and the ponderable c, d hermetries (neutral and charged elementary corpuscles), all indexed by the same integer n . The discrete particle spectrum that the world actually exhibits must be extracted from this pseudo-continuum by a separating term selector. Constructing that separator is the central technical task of Volume II, and it requires first finishing the cosmology that the dimensional theorem of Chapter 2 began.

Six sections develop the programme. Section 5.1 determines the *lower edge* of the complex-hermetry spectrum: a triad of three near-electron masses whose middle term reproduces the empirical electron mass to 0.53%. Section 5.2 introduces the four condensor constants $\{\tau, \mu, \vartheta, \kappa\}$ and uses them to rewrite Heim's four empirical natural constants as algebraic expressions in those four. Section 5.3 states the cosmological problem $A = \dot{D}/D \leq 0$, derives the cosmological master polynomial $aD \approx \tau^{-11/6}$, and points out the paradox $H \gg A$ between the empirical Hubble constant and the structurally allowed expansion rate. Section 5.4 resolves the paradox via Bellert's gravitational-redshift formula and the resulting *optical-universe* segmentation of R_3 . Section 5.5 derives the cosmogonic sphere trinity $\{D_f, D_m, D_p\}$ as the three real roots of the polynomial $\eta^7 - \eta = a$. Section 5.6 closes with the mundal-entelechy principle, the explicit $D(t)$ differential equation, and the numerical estimates for the current age, peak expansion, and total lifetime of the universe.

5.1 The Minimal Complex Condensation and the Lepton Triad

5.1.1 Pseudo-continuum vs. discrete spectrum

The complex-hermetry mass spectrum (4.21) of Chapter 4 contains imponderable terms (gravitons, photons) and ponderable terms (c and d) on the same integer scale. The observable particle spectrum is the discrete sub-set of c and d terms embedded in the dense imaginary pseudo-continuum; a complete theory must produce a *term selector* that strips out the latter.

5.1.2 The metronic projection theorem

Consider an isolated $\mathcal{M}_q$ of mass $m_0 = m(r_0)$ at the lower spectral edge. By Theorem 2.15 its gravitational structure carries the inner radius $s = R_- = eA$ with $A = \frac{3\gamma m_0}{16c^2} = \frac{\tau}{2\lambda}$ (using $E_0 = m_0 c^2 = ch/\lambda$) and the outer realm boundary $R = R_+$ from (2.31). Setting $\zeta^2 = Rs = eRA$ and projecting the spherical level surfaces onto the meridian plane, the projective lattice selector $C = \varepsilon\sqrt{\tau}()$ with projection factor

ε satisfying $\varepsilon^2 \sqrt{F} = s_0 = 1 \text{ [m]}$ yields the following invariant.

Theorem 5.1 (Metronic projection theorem). *For an isolated spherical $\mathcal{M}_q$ source projected on its own gravito-meridian plane,*

$$\boxed{e R \sqrt{\tau} = \varepsilon n F, \quad \varepsilon^2 \sqrt{F} = s_0 = 1 \text{ [m]},} \quad (5.1)$$

in which F is the projected unit area of the level surfaces. For spherical level surfaces $F = \pi E$ with $E = s_0^2$ and $\varepsilon = \pi^{-1/4}$, so (5.1) reduces to $e R \sqrt{\tau} = \pi^{3/4} E n$.

5.1.3 The minimum-mass formula

Theorem 5.2 (Lower bound of the charge- q spectrum). *The smallest mass that can still carry the elementary charge $q \in \mathbb{Z}_{\geq 0}$ satisfies*

$$\boxed{m_q \approx \frac{h}{\beta c} \sqrt[3]{\frac{4\alpha_q}{\varepsilon}}, \quad \beta = \sqrt{\tau \sqrt{\pi}}, \quad \varepsilon \tau = E, \quad \alpha_q = 1 + 4 \left(\frac{q}{\pi} \right)^4.} \quad (5.2)$$

Equivalent form involving only $c, \hbar, \gamma$ and the unit length s_0 :

$$m_q c s_0 \eta_q \sqrt[3]{\eta_q} = 4 \sqrt[4]{\pi} \sqrt[3]{3 \pi \gamma \hbar s_0} \sqrt{\frac{c \hbar}{3 \gamma}}. \quad (5.3)$$

5.1.4 The lepton triad

Theorem 5.3 (Lepton triad). *Identifying $m_0 = m_L$, $m_1 = m_e$ via $m_e \eta \sqrt[3]{\eta} = m_L$, and m_B via $m_B \eta^2 \sqrt[3]{\eta} = m_L$, the lower edge of the complex-hermetry spectrum is the ordered triad*

$$\boxed{m_L < m_e < m_B, \quad s_0 = 1 \text{ [m]}.} \quad (5.4)$$

Numerically:

$$m_L \approx 0.5069 \text{ MeV}, \quad m_e \approx 0.5137 \text{ MeV}, \quad m_B \approx 0.5189 \text{ MeV}. \quad (5.5)$$

m_e agrees with the measured electron rest mass to +0.53%; the deviation is closed by the cellular fine-structure correction discussed at the close of Chapter 4.

Remark 5.1 (Provisional naming). ν_L (mass m_L , neutral, stable) is provisionally called the *lepto-neutrino*; ν_B (mass m_B , neutral, possibly transient) the *bary-neutrino*. These are *not* the empirical β -neutrino of the $n \rightarrow p$ transition, which has mass $m_x \ll m_e$. The names are placeholders until their physical roles are resolved.

5.1.5 The β -process condition

Theorem 5.4 (β -process bound). *Every transition $c \rightleftharpoons d$ between complementary hermetries with the same n proceeds via simultaneous emission of q electrons (and q accompanying β -neutrinos) and is energetically allowed only when*

$$m_c (1 - \eta_q) > q m_e. \quad (5.6)$$

For $q = 1$ this requires $m_c > m_e / (1 - \eta) \approx 101 m_e$, which rules out the speculative $\nu_B \rightarrow e^-$ channel.

5.1.6 Selecting the linear branch of ζ

Remark 5.2 (Why $\lambda' = 2\zeta$ and not $2\pi\zeta$). The derivation of (2.29) of Chapter 2 admitted, in principle, two branches: the planar $\lambda' = 2\zeta$ (taken in the main text) and the circular $\lambda' = 2\pi\zeta$ which would have led to $\pi^2\gamma m_N^3\zeta = h^2$ and a different spectral lower bound. The circular branch produces a d -spectrum edge that does *not* coincide with the empirical electron mass and is therefore excluded. The agreement of (5.5) with m_e confirms the linear branch and, indirectly, a non-trivial coupling between the nuclear forces and the macroscopic gravitational structure.

5.2 Cosmic Actualisation and the Condensor Constants

5.2.1 The unified field tensor on R_6 revisited

For a generic anti-Hermitian field tensor $M_{ik} = -M_{ki}^*$ on R_6 , write

$$\begin{aligned} M_{k5} &= i f_k(x_1, \dots, x_4) \varphi(x_5, x_6), & M_{k6} &= i F_k(x_1, \dots, x_4) \psi(x_5, x_6), \\ M_{56} &= i W(x_5, x_6), & k &= 1, 2, 3, 4, \end{aligned} \quad (5.7)$$

with f, F four-vectors on R_4 and φ, ψ, W scalar functions of the hidden coordinates. The six-divergence-free condition $\overline{\text{div}}_6 M_{ik} = 0$ separates into a real R_4 -current $\mathbf{X}$ and an imaginary R_4 -current $i\mathbf{Z}$:

Proposition 5.5 (Mass and charge currents).

$$\mathbf{X} + f \varphi_{,\varepsilon} + F \psi_{,\eta} + e_5 W_{,\eta} - e_6 W_{,\varepsilon} = \mathbf{0}, \quad (5.8)$$

$$\mathbf{Z} = (e_5 + e_6) (\varphi \text{div}_4 f + \psi \text{div}_4 F). \quad (5.9)$$

$\mathbf{X}$ is the inertial-mass current (always non-zero) and $\mathbf{Z}$ is the electric-charge current (may vanish when f, F are sourceless in the four-divergence sense).

5.2.2 Entelechial and aeonic dimensions

Definition 5.3 (Entelechial and aeonic dimensions). Following a reading of Konrad-Martius:

- $x_5 = i\varepsilon$ is the *entelechial dimension*, parameterising organisational states of an R_4 -structure;
- $x_6 = i\eta$ is the *aeonic dimension*, steering the entelechial structure toward dynamic stability across an R_4 -evolution.

In the macroscopic regime where the second law of thermodynamics holds, the aeonic steering drives entelechy down (entropy up).

Theorem 5.6 (Hidden-coordinate projection). *The R_3 -projection of the entelechial and aeonic data of an elementary corpuscle of mass m is*

$$\underline{\eta} = \frac{h}{2mc}, \quad \underline{\varepsilon} = \frac{h^2}{\gamma m^3}, \quad \boxed{\underline{\eta} = \sqrt[3]{\frac{\tau \underline{\varepsilon}}{3}}}. \quad (5.10)$$

Theorem 5.7 (Aeonic uncertainty). *With the momentum identification $p = h/(2\underline{\eta}) = \pi\hbar/\underline{\eta}$ and the action quantum $w = Nh$, $N \geq 1$,*

$$p \underline{\eta} = \pi\hbar, \quad \frac{w}{2p} \geq \underline{\eta}. \quad (5.11)$$

The lower bound is the aeonic version of the canonical uncertainty relation: any R_4 event is actualised from the aeonic background acausally at the microscopic level; macroscopic causality emerges as the superposition of many such acausal actualisations.

5.2.3 The four condensor constants

Definition 5.4 (Condensor constants). The world manifold carries four irreducible condensor constants:

$$\tau = \frac{\gamma h}{2\omega c^2} \quad (\text{metron, area}), \quad (5.12)$$

$$\mu = \sqrt{\frac{c h}{\gamma}} \quad (\text{maximon / Planck mass}), \quad (5.13)$$

$$\kappa = |\varepsilon_{\pm}| = \frac{3}{\pi^2} \sqrt{\frac{\hbar}{R_-}} \quad (\text{elementary charge}), \quad (5.14)$$

$$\vartheta = \frac{3b}{4c} \tau^{5/6}, \quad b = \frac{e E^{-1/3}}{\sqrt[4]{\pi} \sqrt[3]{2}} \quad (\text{chronon, time}), \quad (5.15)$$

with $R_- = \sqrt{\mu_0/\varepsilon_0}$ the vacuum wave impedance.

Theorem 5.8 (Metrification of the empirical constants). *The four fundamental empirical constants are algebraic expressions in the four condensor constants:*

$h = b \sqrt{\frac{3}{2}} \frac{\mu}{\vartheta} \tau^{4/3}, \quad \gamma = \frac{3}{4} \sqrt{\frac{3}{2}} \left(\frac{b\tau}{\vartheta} \right)^2 \frac{\sqrt[6]{\tau}}{\mu},$ $\varepsilon_0 = (\pi\kappa)^2 \left(\frac{2\pi}{3} \right)^3 \sqrt{\frac{2}{3}} \left(\frac{\vartheta}{b\tau} \right)^2 (\mu \sqrt[6]{\tau})^{-1}, \quad \mu_0 = \frac{6}{\pi^5} \sqrt{\frac{3}{2}} \frac{\mu \sqrt{\tau}}{\kappa^2}.$	(5.16)
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The constant κ appears only in the electromagnetic pair (ε_0, μ_0) ; the mechanical pair (h, γ) depends only on (μ, ϑ, τ) .

5.2.4 Cosmological dependencies

If one further D -dependence $\tau(D)$ is identified (Section 5.3), then the four empirical constants become functions of the universe diameter D :

Corollary 5.9 (Proportionalities under $\tau \rightarrow (aD)^{-6/11}$). *With $\mu, \vartheta, \kappa = \text{const}$,*

$$h \sim D^{-8/11}, \quad \gamma \sim D^{-13/11}, \quad \varepsilon_0 \sim D^{13/11}, \quad \mu_0 \sim D^{-3/11}. \quad (5.17)$$

5.3 The Cosmological Problem and Reference Structures

5.3.1 The cosmological master polynomial

Substituting $L = 1$ in the upper realm boundary $R_+ = D/2$ with the exact lower spectral edge $m_L = m(n_N, 0)$ of (4.21) and the projection theorem (5.1) produces an algebraic relation between the universe diameter D and the metron τ that contains only π , e , and the unit area $E = s_0^2$.

Theorem 5.10 (Cosmological master polynomial).

$$\boxed{f \sqrt{\frac{3}{2}} \left(\frac{D}{4\sqrt{\tau}} f^3 \sqrt{\frac{3}{2}} - 1 \right)^2 = \frac{D}{\sqrt{\tau}}, \quad \left(\frac{eD\sqrt{\tau}}{\pi E} - 1 \right) f^2 = \sqrt{\frac{eD\sqrt{\tau}}{\pi E}}.} \quad (5.18)$$

The asymptotic form valid for $D \gg \sqrt{\tau}$ is

$$aD \approx \tau^{-11/6}, \quad 2a = (3\pi \cdot 2^5)^{-2} E^{-7/3}. \quad (5.19)$$

5.3.2 The cosmological alternative

Definition 5.5 (Cosmological problem). Define the cosmological parameter

$$A := \frac{\dot{D}}{D}, \quad A = 0 \text{ (static)}, \quad A > 0 \text{ (expanding)}, \quad A < 0 \text{ (contracting)}. \quad (5.20)$$

The cosmic helium abundance and the running of the gravito-electric ratio $W = |K_g/K_e|$ between two electrons,

$$3W = 16\pi (a/b)^2 \sqrt[3]{3\tau^2}, \quad (5.21)$$

both favour $A > 0$ in earlier R_4 -strata, with the corresponding $\dot{\tau} < 0$.

Corollary 5.11 (Trend of W and field-line count). With $a^2 N \sqrt[3]{3\tau^2} = 2\pi$ counting the discrete field lines of an isolated electron, the cosmological product

$$3\alpha' N W = (4\pi)^2 \quad (5.22)$$

is a pure cosmological constant determined by the [Wikipedia: Fine-structure_constant](#) alone, independent of $D(t)$.

5.3.3 The cosmological paradox

If the empirical Hubble redshift $z \approx Bs$ is interpreted as a recession velocity Doppler effect $v \approx Hs$ with $H = cB$, then the redshift of an emission line over a 10^9 -year baseline *should* satisfy $11z = -H\delta T$, yielding a cosmological blue-shift of $\approx 0.7\%$ — directly contrary to the observed red-shift.

Theorem 5.12 (Cosmological paradox). The structural expansion rate A derived from the master polynomial (5.18) satisfies

$$A \ll H, \quad z \approx Bs, \quad H = cB, \quad (5.23)$$

i.e. the Hubble redshift cannot be a kinematic Doppler effect of the cosmic expansion in this framework. Some other mechanism must produce z .

5.3.4 Three photon-wavelength components

Proposition 5.13 (Photon wavelength variation). *The total cosmic shift of a photon wavelength has three components,*

$$c z_D = A s, \quad z_S = -\frac{39 \lambda}{11 c} A, \quad z_a = \frac{16 \tau}{9 \lambda^2}, \quad (5.24)$$

summing to $z_k = z_D + z_S + z_a$. All three lie below the spectroscopic measurement threshold; none can account for the empirical Hubble redshift.

5.3.5 Reference photon

Theorem 5.14 (Reference wavelength). *The wavelength at which $z_S + z_a = 0$ — i.e. at which the structural blue-shift exactly compensates the light-aging red-shift — is*

$$\lambda_r^3 \approx \frac{c \tau}{2 A}, \quad 4 \lambda_r^3 \approx \tau D, \quad c \lambda_r^3 = \text{const.} \quad (5.25)$$

Photons of wavelength λ_r serve as cosmological reference structures.

5.4 Resolution of the Cosmological Paradox

5.4.1 Gravitational latency outside the cutoff

Postulate 5.6 (Latent gravity outside the cutoff). The repulsive branch $g = \nabla \varphi > 0$ of the field (2.27) at $r > \zeta$ is *latent*: it exerts no ponderomotive force in free outward motion, but a mass approaching the source from $r > \zeta$ is decelerated by it. Photons traversing such a region therefore lose field-energy as they cross.

5.4.2 Bellert's redshift formula

Tracking the field-energy loss of a photon passing through a cylindrical column of mass density σ aligned with the photon trajectory, one obtains:

Theorem 5.15 (Bellert formula). *The cosmological redshift of a distant emission line at proper distance $s \gg \zeta$ is*

$$z \approx \frac{B s}{1 - B s}, \quad c B = \sqrt{\pi e \gamma \sigma}, \quad (5.26)$$

with σ the integrated mass density along the line of sight. For $B s \ll 1$ the linear approximation $z \approx B s$ matches Hubble's law with $H = c B$.

Remark 5.7 (Historical attribution). The same formula (5.26) was derived independently and without an underlying theory by S. Bellert (1969); the present derivation grounds it in Postulate 5.6 and the modified Newton potential of Theorem 2.15.

5.4.3 Optical universes

Theorem 5.16 (Optical horizon and universe diameter). *For $s = y$ the Bellert formula diverges, defining an optical horizon:*

$$\boxed{y \sqrt{\pi e \gamma \sigma} \approx c, \quad 2y \ll D.} \quad (5.27)$$

The physical universe of diameter D is tiled into a large number of optical universes of radius $y < \infty$. The Hubble radius is y , not D .

Proposition 5.17 (Vacuoles of red-shift). *Within each gravitational reach $r \leq \zeta$ of an attracting mass system $\sigma = 0$, so the static red-shift vanishes: $z = 0$ for $s \leq \zeta$, $z \rightarrow +\infty$ as $s \rightarrow y$, with red-shift growth confined to $\zeta < s < y$.*

Remark 5.8 (Andromeda discriminator). The Andromeda galaxy lies within the Milky Way's vacuole (well inside ζ); its observed spectral red-shift cannot be a $v = Hs$ recession in this framework, yet it is observed. The column-integrated σ along the line of sight outside the vacuole accounts for it, confirming Postulate 5.6.

5.4.4 Numerical values

Three empirical estimates of intergalactic mass density yield three Hubble values:

$\sigma \text{ [kg m}^{-3}\text{]}$	$H \text{ [s}^{-1}\text{]}$	$y \text{ [ly]}$
3×10^{-28}	4.20×10^{-19}	7.54×10^{10}
1×10^{-26}	2.39×10^{-18}	1.33×10^{10}
2×10^{-26}	3.38×10^{-18}	9.39×10^9

(5.28)

agreeing with Heidmann's (1980) empirical $H = (80 \pm 21) \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Wikipedia: Hubble%27s_law).

5.5 Cosmogenesis and the Sphere Trinity

5.5.1 Backwards integration to the metronic origin

By Theorem 5.10 and the sign $\dot{\tau} < 0$ in the expansion phase, going backwards in time $\tau(t)$ grows while $D(t)$ shrinks. At a finite past time $t = 0$ the metron reaches a value such that a single metron τ_0 encloses the entire R_3 :

$$\boxed{\tau_0 = \pi D_0^2.} \quad (5.29)$$

Before $t = 0$ the metric concept fails (*apeiron*).

5.5.2 The cosmogonic polynomial

Theorem 5.18 (Cosmogonic polynomial). *Substituting (5.29) into the master polynomial yields*

$$\boxed{\eta^7 - \eta = \pm a, \quad 2\eta^2 = f_{(0)} \sqrt[6]{6/\pi}, \quad a\sqrt{\pi} = \sqrt[6]{\pi/6},} \quad (5.30)$$

with $f_{(0)}$ determined implicitly by $f_{(0)} (e D_0^2 / (E\sqrt{\pi}) - 1)^{1/2} = \sqrt[4]{e D_0^2 / (E\sqrt{\pi})}$.

5.5.3 Three real roots

Proposition 5.19 (Root structure of $\eta^7 - \eta = a$). *The polynomial $y(\eta) = \eta^7 - \eta - a$ with $a > 0$ has stationary points $\eta_{\pm} = \pm \sqrt[6]{1/7}$ and inflection $\eta_w = 0$, with $y_{\max} > 0$, $y_w = -a$, $y_{\min} < -a$. It therefore has three real roots*

$$-1 < \eta_1 < -\sqrt[6]{1/7} < \eta_2 < 0 < 1 < \eta_3, \quad (5.31)$$

plus two pairs of complex-conjugate roots, the latter eliminated by the positivity requirement on $f_{(0)} = 2\sqrt[6]{\pi/6}\eta^2 > 0$.

5.5.4 The cosmogonic sphere trinity

Theorem 5.20 (Sphere trinity at $t = 0$). *The three real roots of (5.30) give three real positive diameters $\{D_1, D_2, D_3\}$ in the ordering $D_3 < D_1 < D_2$. Identifying these as fundamental, meso, and proto sphere respectively,*

$$\boxed{D_f < D_m < D_p}, \quad (5.32)$$

numerically

$$D_f = 0.910 \text{ m}, \quad D_m = 1.064 \text{ m}, \quad D_p = 3.701 \text{ m}. \quad (5.33)$$

5.5.5 End-time trinity and the antiworld

Theorem 5.21 (End-time mirror trinity). *The negative branch of (5.30) yields a mirror trinity $T_E = D_{fmp}^{yx-}$ in which the fundamental and meso-sphere semantics are exchanged with respect to the cosmogonic trinity $T_K = D_{fmp}^{xy+}$. The exchange encodes the antiparallel time orientation of the anti-world R_4^- versus the world R_4^+ :*

$$T_K^+ \xrightarrow[\cos \alpha_4 > 0]{D(R_3^+)} T_E^+ \equiv T_K^- \xrightarrow[\cos \alpha_4 < 0]{D(R_3^-)} T_E^- \equiv T_K^+. \quad (5.34)$$

5.5.6 Three-step actualisation

Remark 5.9 (Chronon-spaced actualisation). Cosmic actualisation proceeds in three chronon-separated steps: the proto-sphere enters time first, its meso-sphere image becomes the new meso-sphere ϑ later, and the fundamental-sphere image follows another ϑ later. The present always contains a triple $\{p, m, f\}$ of metrons separated by ϑ ; the world “present” is a finite time interval of length 2ϑ , not a zero-measure event.

5.6 Boundaries of Space and Time

5.6.1 The mundal entelechy principle

Postulate 5.10 (Mundal entelechy). Integrated over a momentary R_5 -hypersurface,

$$\boxed{\mathbf{x}_5^{(g)} + \mathbf{x}_6^{(g)} = i\mathbf{M} = \text{const}(x_4) < \infty, \quad \dot{\mathbf{x}}_g^2 + \dot{\mathbf{y}}_g^2 = 0.} \quad (5.35)$$

The integral hidden-coordinate content is conserved along the cosmic evolution. Locally the projection content fluctuates and drives the actualisation.

5.6.2 The cosmic-evolution differential equation

Theorem 5.22 (Global $D(t)$ equation). *Combining the integral relation $\dot{D} = 2c$ (which follows from (5.35) via $w_g = c$) with the D -dependence $c = C D^a$, $11a = -5$, of Corollary 5.9 yields*

$$\boxed{D \ddot{D} - a \dot{D}^2 = 0, \quad 11a = -5,} \quad (5.36)$$

with first integral $D^{-a} \dot{D} = 2C = \text{const}$ and second integral

$$D^\beta - D_1^\beta = 2\beta C(t - t_1), \quad \beta = 1 - a = \frac{16}{11}. \quad (5.37)$$

5.6.3 Maximum expansion and the lifetime of the universe

Theorem 5.23 (Lifetime parameters). *The peak-expansion ratio derived from the spectral lower bound is*

$$\left(\frac{\Lambda}{D}\right)^{3/11} \sqrt{E} = \frac{2^9 e \sqrt[4]{\pi}}{9 \pi^2 \sqrt[3]{2}} \left(\frac{\tau}{E}\right)^{11/6} D. \quad (5.38)$$

Inserting the present-day values $D_T \approx 6.03 \times 10^{125} \text{ m}$ and $\tau_T \approx 6.15 \times 10^{-70} \text{ m}^2$ yields

$$\boxed{T \approx 1.72 \times 10^{115} \text{ s}, \quad T_\Lambda \approx 2.75 \times 10^{117} \text{ s}, \quad \Theta = 2 T_\Lambda \approx 320 T,} \quad (5.39)$$

with $\Lambda \approx 1.56 \times 10^{126} \text{ m} \approx 2.6 D_T$.

Corollary 5.24 (Matter cosmogenesis cannot coincide with $t = 0$). *If matter creation had occurred at $t = 0$, the stellar thermonuclear processes operating over $\sim 10^{115}$ seconds would have consumed all hydrogen, contradicting the present cosmic abundance. The material cosmogenesis must therefore have occurred at some $T_M \gg 0$ with $T - T_M \ll T$.*

5.6.4 The closing synmetronic problem

The cosmogonic sphere trinity has $\omega = 3$ as the count of distinguishable mono-metric spheres; via (2.34) this is also the maximum number of admissible lattice kernels ${}^2\bar{\kappa}_{(\mu)}$ of the polymetric correlator $\hat{\gamma} = ({}^2\hat{\gamma}_{(\mu\nu)})_3$. The cosmogonic trinity therefore provides the missing existence proof that the heuristic ansatz of polymetric composition is actually realised in nature.

Theorem 5.25 (Synmetronic ansatz). *There exist three lattice-kernel selectors of the world manifold*

$$\boxed{{}^2\bar{\kappa}_{(1)} = {}^2\bar{\kappa}_{(1)}(C_5, C_6), \quad {}^2\bar{\kappa}_{(2)} = {}^2\bar{\kappa}_{(2)}(C_4), \quad {}^2\bar{\kappa}_{(3)} = {}^2\bar{\kappa}_{(3)}(C_1, C_2, C_3),} \quad (5.40)$$

each non-Hermitian, with Gram products ${}^2\tilde{\gamma}_{(\mu\nu)} = \text{sp}({}^2\tilde{\kappa}_{(\mu)} \times {}^2\tilde{\kappa}_{(\nu)})$ assembling into the polymetric correlator $\hat{\gamma} = ({}^2\tilde{\gamma}_{(\mu\nu)})_3$. The four hermetry forms $H(\cdot)$ of Theorem 4.4 are recovered by applying sieve operators to $\hat{\gamma}$:

$$\hat{\gamma}_a = \hat{\gamma}_a(C_5, C_6), \quad \hat{\gamma}_b = \hat{\gamma}_b(C_4, C_5, C_6), \quad \hat{\gamma}_c = \hat{\gamma}_c(C_{1\dots 3}, C_5, C_6), \quad \hat{\gamma}_d = \hat{\gamma}_d(C_{1\dots 6}). \quad (5.41)$$

Remark 5.11 (Closure of Volume I). The synmetronic ansatz (5.40) closes the heuristic conjecture (2.26) of Section 2.3 and gives the existence proof for the three partial lattice kernels demanded by the dimensional axiom $M = N/p = 3$. The technical exploitation of this ansatz — applying the discrete cellular machinery of Chapter 3 to split the pseudo-continuum from the discrete particle spectrum — is the task of the following chapter.

Summary of notation introduced in this chapter

Symbol	Object
$A = \dot{D}/D$	Cosmological expansion parameter
$D(t), \Lambda$	Universe diameter and maximum expansion
D_f, D_m, D_p	Fundamental / meso / proto sphere
T_K, T_E	Cosmogonic and end-time sphere trinity
$\tau_0 = \pi D_0^2$	Initial monometronic enclosure
m_L, m_e, m_B	Lepto-neutrino, electron, bary-neutrino
$\alpha_q = 1 + 4(q/\pi)^4$	Charge-dependent mass coefficient
$\eta_q, \eta \equiv \eta_1$	Charge-correction factor
$\mathbf{X}, \mathbf{Z}$	Inertial-mass and electric-charge R_4 -currents
φ, ψ, W	Hidden-coordinate field components
$\underline{\varepsilon}, \underline{\eta}$	R_3 -projections of the hidden coordinates
ϑ	Chronon (elementary time interval)
$\mu = \sqrt{ch/\gamma}$	Maximon / Planck mass
$\kappa = \varepsilon_{\pm} $	Elementary charge
b	Geometric prefactor in ϑ
$B = H/c$	Bellert (Hubble) constant
y	Optical-universe (Hubble) radius
σ	Column mass density
λ_r	Reference photon wavelength
z_D, z_S, z_a	Dynamical / structural / aging redshift components
T, T_{Λ}, Θ	Present age, peak-expansion time, total life ($\Theta = 2T_{\Lambda}$)
${}^2\tilde{\kappa}_{(\mu)}$	Lattice kernel for partial structure μ
$\hat{\gamma}$	Polymetric correlator $({}^2\tilde{\gamma}_{(\mu\nu)})_3$

Synmetronics of the World

Scope of the chapter. Chapter 5 closed with the *synmetronic ansatz* (5.40): three non-Hermitian lattice-kernel selectors ${}^2\bar{\kappa}_{(1)}, {}^2\bar{\kappa}_{(2)}, {}^2\bar{\kappa}_{(3)}$ of the world manifold, each tied to a distinct subset of world coordinates, whose pairwise Gram products assemble into a 3×3 polymetric correlator $\hat{\gamma} = ({}^2\tilde{\gamma}_{(\mu\nu)})_3$. The technical task of the present chapter is to develop the discrete calculus of that correlator and to derive the *synmetronic world selector equation* — the operator equation whose eigenvalue spectrum decomposes the pseudo-continuum of Chapter 5 into the discrete particle spectrum and the imponderable background, and whose solutions classify into the four hermetry forms of Theorem 4.4.

Five sections develop the construction. Section 6.1 sets up the structure-unit interpretation of the lattice kernels, gives the explicit 3×3 correlator matrix of each hermetry form, and introduces the *proto-field / field-activator* decomposition. Section 6.2 identifies every antihermitian metric-component with a discrete-rotor spin-field selector, deriving the universal covariant Hermiticity of all fundamental condensers. Section 6.3 states and analyses the *synmetronic fundamental problem* (6.16), proves the existence of the metronic Hilbert space for its eigenvalue spectrum, derives the spectral selection rule $Z = \omega^2 q(2\omega - q)$, and establishes the spin-compensation principle that governs every composition of partial condensation stages. Section 6.4 solves the system for each of the four hermetry forms a, b, c, d , producing the explicit form of the coupling selector $Q_{(\mu\nu)}^{(\kappa\lambda)}$ and the *coupling-extremum* schema $(\frac{\kappa\lambda}{\mu\nu})$. Section 6.5 closes with the *condensor flux*: a variational principle on the structure compressor that drives every metronic compression toward its energetic minimum, together with the universal conservation law of the compositional field.

6.1 Structure Units, Lattice Kernels, and the Correlator

6.1.1 Structure-unit interpretation of the lattice kernels

By the *synmetronic ansatz* (5.40) of Chapter 5, the world manifold R_6 supports three lattice kernels

$${}^2\bar{\kappa}_{(1)} = {}^2\bar{\kappa}_{(1)}(C_5, C_6), \quad {}^2\bar{\kappa}_{(2)} = {}^2\bar{\kappa}_{(2)}(C_4), \quad {}^2\bar{\kappa}_{(3)} = {}^2\bar{\kappa}_{(3)}(C_1, C_2, C_3), \quad (6.1)$$

which we interpret as three independent *structure units*: the trans-coordinate pair (x_5, x_6) , the time coordinate x_4 , and the physical R_3 respectively.

Definition 6.1 (Polymetric correlator). The polymetric correlator of the world manifold is the non-Hermitian 3×3 Hyper-matrix

$$\hat{\gamma} = ({}^2\tilde{\gamma}_{(\mu\nu)})_3, \quad {}^2\tilde{\gamma}_{(\mu\nu)} = \text{sp}({}^2\bar{\kappa}_{(\mu)} \times {}^2\bar{\kappa}_{(\nu)}). \quad (6.2)$$

Its diagonal entries ${}^2\tilde{\gamma}_{(\nu\nu)}$ are the partial metric selectors of the three partial structures; its $\omega(\omega - 1) = 6$ off-diagonals are the *correlation mediators* which couple distinct partial structures.

Proposition 6.1 (Hermitian/anti-Hermitian split). *Every kernel decomposes as ${}^2\bar{\kappa}_{(v)} = {}^2\bar{\kappa}_{(v)+} + {}^2\bar{\kappa}_{(v)-}$ with ${}^2\bar{\kappa}_{(v)+} = {}^2\bar{\kappa}_{(v)+}^\times$, ${}^2\bar{\kappa}_{(v)-} = -{}^2\bar{\kappa}_{(v)-}^\times$, and the correlator inherits this split: ${}^2\tilde{\gamma}_{(\mu\nu)} = {}^2\tilde{\gamma}_{(\mu\nu)+} + {}^2\tilde{\gamma}_{(\mu\nu)-}$. While the composite field ${}^2\tilde{\gamma}_{()} = {}^2\tilde{\gamma}$ is itself Hermitian, none of the individual entries ${}^2\tilde{\gamma}_{(\mu\nu)}$ are forced to be.*

6.1.2 The hermetry-form matrices

Applying sieve operators $S(\mu)$ that switch off the chosen structure unit, each of the four hermetry forms produces a different 3×3 correlator schema. With ${}^2\bar{E}$ for the unit selector and $\hat{0}$ for the zero selector:

Theorem 6.2 (Hermety-form correlators). *The correlator of each hermetry form has the explicit shape*

$$\hat{\gamma}_a = S(2,3) \hat{\gamma} = \begin{pmatrix} {}^2\tilde{\gamma}_{(11)} & {}^2\bar{\kappa}_{(1)} & {}^2\bar{\kappa}_{(1)} \\ {}^2\bar{\kappa}_{(1)} & {}^2\bar{E} & {}^2\bar{E} \\ {}^2\bar{\kappa}_{(1)} & {}^2\bar{E} & {}^2\bar{E} \end{pmatrix} = \text{pseudo-bimetric (self-condensation),} \quad (6.3)$$

$$\hat{\gamma}_b = S(3) \hat{\gamma} = \begin{pmatrix} {}^2\tilde{\gamma}_{(11)} & {}^2\tilde{\gamma}_{(12)} & {}^2\bar{\kappa}_{(1)} \\ {}^2\tilde{\gamma}_{(21)} & {}^2\tilde{\gamma}_{(22)} & {}^2\bar{\kappa}_{(2)} \\ {}^2\bar{\kappa}_{(1)} & {}^2\bar{\kappa}_{(2)} & {}^2\bar{E} \end{pmatrix} = \text{time-like pseudo-hexametric,} \quad (6.4)$$

$$\hat{\gamma}_c = S(2) \hat{\gamma} = \begin{pmatrix} {}^2\tilde{\gamma}_{(11)} & {}^2\bar{\kappa}_{(1)} & {}^2\tilde{\gamma}_{(13)} \\ {}^2\bar{\kappa}_{(1)} & {}^2\bar{E} & {}^2\bar{\kappa}_{(3)} \\ {}^2\tilde{\gamma}_{(31)} & {}^2\bar{\kappa}_{(3)} & {}^2\tilde{\gamma}_{(33)} \end{pmatrix} = \text{space-like pseudo-hexametric,} \quad (6.5)$$

$$\hat{\gamma}_d = ({}^2\tilde{\gamma}_{(\mu\nu)})_3 = \text{total enneametric (space-time condensation).} \quad (6.6)$$

The forms a, b are imaginary (built only from ${}^2\bar{\kappa}_{(1)}, {}^2\bar{\kappa}_{(2)}$); the forms c, d become complex through the inclusion of ${}^2\bar{\kappa}_{(3)}$.

6.1.3 Proto-field and field activator

If two consecutive sieves reduce the correlator to a single diagonal entry, the Hermitian condition on the composite field forces the residual to be entirely Hermitian:

Theorem 6.3 (Proto-field interpretation). *Under double sieve action $S(i)S(k){}^2\tilde{\gamma}$, $i \neq k \neq v$, the composite field reduces to the single Hermitian remainder*

$${}^2\tilde{\gamma}({}^2\tilde{\gamma}_{(vv)}) = {}^2\tilde{\gamma}_{(vv)+} \quad \text{with} \quad {}^2\tilde{\gamma}_{(vv)-} = {}^2\bar{0}. \quad (6.7)$$

The Hermitian remainder ${}^2\tilde{\gamma}_{(vv)+}$ is an un-quantised proto-field: no eigenvalue spectrum exists in the single-partial-structure regime. The antihermitian residual ${}^2\tilde{\gamma}_{(vv)-}$, present in the full correlator, is therefore the field activator that promotes a proto-field into a correlation. Discrete condensation stages emerge only when two activated proto-fields enter into correlation.

Corollary 6.4 (Universal extension of the names). *By extension, in any correlator entry ${}^2\tilde{\gamma}_{(\mu\nu)}$:*

- the Hermitian part ${}^2\tilde{\gamma}_{(\mu\nu)+}$ is the proto-selector;
- the antihermitian part ${}^2\tilde{\gamma}_{(\mu\nu)-}$ is the activation selector.

Activation always proceeds through the unique real proto-field of form a ; every transmutation among b, c, d in R_3 is mediated by a graviton system arising from the a -channel self-condensations.

6.1.4 Composition laws

The synmetronic ansatz extends from the correlator to every derived selector:

Proposition 6.5 (Composition laws). *The composition rules (2.26) extend to*

$$\begin{aligned} {}^2\bar{\kappa}({}^2\bar{\kappa}_{(\mu)})_1^3 &= {}^2\bar{\kappa}^\times, & \bar{\psi}(\bar{\psi}_{(\mu)})_1^3 &= \bar{\psi}^\times, \\ \widehat{\{\}} &= f\left(\left[\frac{\kappa\lambda}{\mu\nu}\right]\right)_1^3 = \widehat{\{\}}^\times, & {}^2\bar{\kappa}_{(\mu)} &\neq {}^2\bar{\kappa}_{(\mu)}^\times, \quad \left[\lambda_\mu^\kappa\right]_{(\neq)}^{(\nu)} \left[\lambda_\mu^\kappa\right]_{(\neq)}^{(\nu)} \times. \end{aligned} \quad (6.8)$$

The non-Hermiticity of the partial fundamental condensors is carried only on the covariant basis signature, never on the contra-variant or interaction signature.

6.1.5 Infinitesimal form

The infinitesimal limit of the partial condensor admits the closed form

$$2\{ikl\}_{(ab)} = (\text{sp}({}^2\bar{a} \times \text{rot}_6 {}^2\bar{b}))_{kli} + (\text{sp}((\text{rot}_6 {}^2\bar{a}) \times {}^2\bar{b}))_{kil} + (\text{sp}({}^2\bar{a} \times \widehat{\text{div}}_6 {}^2\bar{b}))_{ikl} + (\text{sp}((\widehat{\text{div}}_6 {}^2\bar{a}) \times {}^2\bar{b}))_{lki}, \quad (6.9)$$

in which the quadratic metron differentials of the discrete calculus have collapsed to zero. This is the $\tau \rightarrow 0$ form of $\{\}; n$.

6.2 Spin-Field Activation

6.2.1 Antihermetric structure and proto-field constancy

Let V_r be the hermetric subspace and V_s its antihermetric complement, $r + s = 6$. The hermetric components of the proto-field are constant along antihermetric directions; the antihermetric components of the activator are not. Explicitly:

Theorem 6.6 (Proto-field and activator structure). *Within the antihermetric subspace,*

$$\partial_{\bar{k}} \gamma_{+il}^{(pq)} = 0, \quad \partial_{\bar{k}} \gamma_{-il}^{(pq)} = \frac{\alpha_{\bar{k}}}{\alpha_i} \partial_i \gamma_{\bar{k}l}^{(pq)} - \frac{\alpha_{\bar{k}}}{\alpha_l} \partial_l \gamma_{\bar{k}i}^{(pq)}. \quad (6.10)$$

The proto-field is constant on the antihermetric subspace; the activator is a non-trivial cochain there.

6.2.2 The activator is a discrete spin rotor

Theorem 6.7 (Activator = spin rotor). *The antihermitian metric component admits, on the antihermetric subspace, the metron-integrated form*

$${}^2\tilde{\gamma}_{(pq)-} = \text{ROT}_{(x)}\bar{A}_{(pq)} = \frac{1}{\tau}\text{ROT}_{(x)}\bar{\phi}_{(pq)} = {}^2\bar{P}_{(pq)}, \quad (6.11)$$

i.e. every activator is the discrete cellular rotor of an underlying spin-field selector $\bar{\phi}_{(pq)}$, with the constant of proportionality fixed by the metron $\tau = \text{const.}$ In the hermetric directions m the proto-field components carry the same spin field through

$$\tau \alpha_m \gamma_{+\bar{k}m}^{(pq)} = \partial_m \phi_{\bar{k}}^{(pq)}. \quad (6.12)$$

Remark 6.2 (Universality). Equation (6.11) was derived under the antihermetric ansatz but in fact holds in every regime: the antihermitian part of every correlator entry is the discrete spin rotor of an underlying $\bar{\phi}_{(pq)}$. The activators carry the spin orientation of the metronic substrate; the proto-fields carry the structural cells. Activators vanish identically on antihermetric indices: the spin-rotor lifts only on hermetric components.

6.2.3 Universal covariant Hermiticity

A direct consequence of Theorems 6.6–6.7 is that every partial fundamental condensor inherits a *covariant Hermitian* character on its basis signature, irrespective of contra-signature behaviour:

Theorem 6.8 (Universal covariant Hermiticity). *Every partial fundamental condensor of every hermetry form satisfies*

$$\left[\begin{smallmatrix} \kappa \\ \lambda \end{smallmatrix} \mu \right]_{(\cdot)}^{(\nu)} = \left[\begin{smallmatrix} \widehat{\kappa\lambda} \\ + \\ \mu \end{smallmatrix} \right]_{+} = \left[\begin{smallmatrix} \kappa \\ \lambda \end{smallmatrix} \mu \right]_{(\cdot)}^{(\nu)} \times. \quad (6.13)$$

In particular, the antihermitian part of every fundamental condensor vanishes: $\left[\begin{smallmatrix} \widehat{\kappa\lambda} \\ + \\ \mu \end{smallmatrix} \right]_{-} = \hat{0}$.

6.2.4 Wechselwirkung vs Korrelation: terminological refinement

Definition 6.3 (Correlation vs. correspondence). On the world manifold R_6 we adopt the following refinement of the general term *interaction*:

- a *correlation* is an interaction internal to the correlator $\hat{\gamma}$ — i.e. a relation between two of its entries ${}^2\tilde{\gamma}_{(\mu\nu)}$;
- a *correspondence* is an interaction external to a condensation, mediated by the world condensor $\widehat{\{\}}\}$ and relating two distinct condensation eigenvalues $\bar{\lambda}$.

6.3 The Symmetronic Fundamental Problem

6.3.1 The symmetronic compositional law

By the universal Hermiticity (6.13), the four-index sum over all signatures reduces to its symmetric part:

Theorem 6.9 (Synmetronic composition of the world condensor). *The total world condensor is the bare sum*

$$\widehat{\{\}} = \sum_{\varepsilon} \left(\left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_+ + \text{sp}^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times \left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_+ \right), \quad \varepsilon \equiv (\kappa, \lambda, \mu, \nu), \quad (6.14)$$

with ${}^2\bar{Q}_{(\mu\nu)}^{(\kappa\lambda)}$ the coupling cochain of (3.45), restricted to act between distinct correlator entries.

6.3.2 The synmetronic world selector equation

Substituting (6.14) into the world selector $L\widehat{\{\}} = {}^4\bar{0}$ of Theorem 4.2, expanding the resulting structure compressor in the polymetric basis, and demanding that the result hold signature by signature, one obtains:

Theorem 6.10 (Synmetronic world selector). *The synmetronic world selector*

$$L_{(\mu\nu)}^{(\kappa\lambda)} = \Lambda_{(\mu\nu)}^{(\kappa\lambda)} - \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \times (1 + \text{sp}^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times ()) \quad (6.15)$$

satisfies, on every signature quadruple $(\kappa, \lambda, \mu, \nu)$,

$$L_{(\mu\nu)}^{(\kappa\lambda)} ; \left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_+ = {}^4\bar{0}, \quad (6.16)$$

with $\Lambda_{(\mu\nu)}^{(\kappa\lambda)} = K_{(\mu\nu)}^{(\kappa\lambda)} ; (1 + \text{sp}^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times ()) + D_{(\mu\nu)}^{(\kappa\lambda)}$, $K_{(\mu\nu)}^{(\kappa\lambda)}$ the synmetronic Bianchi order-raising selector, and $D_{(\mu\nu)}^{(\kappa\lambda)}$ a quadratic coupling selector.

6.3.3 Spin-compensation principle

The eigenvalue decomposition of (6.16) on each signature yields the central physical principle of the synmetronic construction:

Theorem 6.11 (Spin compensation in composition). *The total condensation eigenvalue $\bar{\lambda}$ of the composite field decomposes as a δ -graded sum over signatures,*

$$\bar{\lambda} \widehat{\{\}} = \sum_{\varepsilon} \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_+^*, \quad \left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_+^* = (1 + \text{sp}^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times ()); \left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_+. \quad (6.17)$$

The composite field carries no spin field; every spin contribution from the partial structures must therefore cancel. Whenever a stationary correlation is perturbed so that a spin field appears, a compensating spin field of another partial structure must be induced; this induction activates the partner proto-field and drives the transmutation of the composite hermetry form.

6.3.4 Existence of the metronic Hilbert space

Theorem 6.12 (Metronic function space). *The eigenvalue equation (6.16) carries a well-defined metronic Hilbert space iff*

$$\left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_- = \hat{0}, \quad \int_{\Omega} \left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_+ ; () \left[\frac{\widehat{\kappa\lambda}}{\mu\nu} \right]_+^{\times} ; () d\Omega < \infty, \quad (6.18)$$

together with the analogous convergence of the coupling cochain $\text{sp}({}^2\bar{Q}_{(\mu\nu)}^{(\kappa\lambda)}) \times [\lambda_\mu^\kappa]_{(\cdot)}^{(\nu)}$. Both conditions are automatically guaranteed by the universal covariant Hermiticity (6.13), so the metronic function space always exists for every solution of (6.16).

6.3.5 Spectral selection rule

In an abstract R_n with $\omega = n/2$ partial structures and $q \leq \omega$ hermetric structure units, only a subset of the ω^4 fundamental condensors carry non-trivial content:

Theorem 6.13 (Spectral selection rule). *Under the antihermetric ansatz, the number of non-trivial fundamental condensors of (6.16) is*

$$\boxed{Z = \omega^2 q (2\omega - q), \quad 0 \leq q \leq \omega.} \quad (6.19)$$

The $\omega^2(\omega - q)^2$ vanishing condensors are exactly those whose contra-signature points into an antihermetric structure unit. For R_6 with $\omega = 3$, the maximal value $Z(q = 3) = 81$ confirms the form-d totality.

Remark 6.4 (Reduction by indices). In addition to (6.19), further vanishing arises when the covariant basis index points into an antihermetric structure unit: $\lambda_{(\kappa\lambda)m}^{(\mu\nu)}(\tilde{k}, l) [\lambda_\mu^\kappa]_{(=)}^{(\nu)} = 0$. Equivalently $[\tilde{k}l]_{(\kappa\lambda)} = 0$ on those entries.

6.3.6 Two matrix traces of the symmetronic equation

Proposition 6.14 (Matrix traces). *Equation (6.16) admits two non-trivial traces*

$$\text{sp } L_{(\mu\nu)}^{(\kappa\lambda)} ; [\lambda_\mu^\kappa]_{(\cdot)}^{(\nu)} = {}^2\bar{0}, \quad L_{(\mu\nu)}^{(\kappa\lambda)} ; \text{sp } [\lambda_\mu^\kappa]_{(\cdot)}^{(\nu)} = {}^2\bar{0}, \quad (6.20)$$

the former yielding the symmetronic Einstein analogue, the latter the symmetronic antisymmetric companion.

6.4 Symmetronics of the Hermetry Forms

6.4.1 Closed form of the coupling selector

In the second metronic validity regime (high cell counts, $\Lambda_{kl} \rightarrow 0$) of Section 4.2, the coupling selector admits an explicit closed form. With $\alpha_l^{(\kappa\lambda)}$ the correlation exponent of the $(\kappa\lambda)$ signature and $\psi_{ll}^{(\kappa\lambda)} = \psi_{ll}^{\alpha_l^{(\kappa\lambda)}} :$

Theorem 6.15 (Coupling selector).

$$\boxed{Q_{(\mu\nu)}^{(\kappa\lambda)} = A_{(\mu\nu)}^{(\kappa\lambda)} \sum_{l=1}^q C_{(\mu\nu)l}^{(\kappa\lambda)} (E - e^{-\lambda_l \mu})^{-\alpha_l} \cdot \frac{|\delta_{kl}(e^{\lambda_l^{(\kappa\lambda)} \mu} - E)^{\alpha_{kl}^{(\kappa\lambda)}}|_q}{|\delta_{kl}(e^{\lambda_l^{(\mu\nu)} \mu} - E)^{\alpha_{kl}^{(\mu\nu)}}|_q} - q E.} \quad (6.21)$$

The function depends only on the linear combination $\mu = \sum_k \alpha_k(\cdot)_k$ of hermetric lattice selectors, and its course is parameterised by $N = 8q + 1$ correlation indices including the three eigenvalue spectra $\lambda_l, \lambda_l^{(\mu\nu)}, \lambda_l^{(\kappa\lambda)}$.

6.4.2 Coupling extrema and the coupling group

Theorem 6.16 (Coupling extrema). *The stationary points of $Q_{(\mu\nu)}^{(\kappa\lambda)}$ are the coupling extrema, simultaneously characterised by*

$$\boxed{\partial Q_{(\mu\nu)}^{(\kappa\lambda)} = 0, \quad \partial \psi_{ll} = 0, \quad \partial^2 {}^2\tilde{\gamma}_{(\kappa\lambda)} = {}^2\bar{0}, \quad \partial^2 {}^2\tilde{\gamma}_{(\mu\nu)} = {}^2\bar{0}.}$$

(6.22)

At each extremum, the coupling partners enter a pseudo-antihermitic configuration in which the hyper-selectors of the participating partial structures align with the lattice selectors; the proto-fields enter their inflection bands while the composite eigenvalues ψ_{ll} realise local plateaux.

Definition 6.5 (Coupling group). The set of all coupling extrema obtainable by permuting the signature indices of a fixed quadruple (keeping each $\{\kappa, \lambda\}$ and $\{\mu, \nu\}$ pair together) forms the *coupling group*

$$Q_{(\mu\nu)\text{ext}}^{(\kappa\lambda)} \equiv \begin{pmatrix} \kappa & \lambda \\ \mu & \nu \end{pmatrix}. \quad (6.23)$$

A coupling group contains at most 8 coupling selectors.

6.4.3 Brücken, Quellen, Senken

Remark 6.6 (Coupling-graph language). Around a coupling group, the remaining condensers organise into

- *primary pseudo-antihermitian* condensers whose basis is the coupling group's signature pair;
- *secondary pseudo-antihermitian* condensers whose basis contains one index of the coupling group;
- *primary hermitic* and *secondary hermitic* condensers whose basis avoids the coupling group.

Between two coupling groups, identical condensers of either type can serve as *bridges* (when they appear with the same character in both groups); identical condensers that switch from hermitic in one group to pseudo-antihermitian in the other serve as *sources* (loss-side) or *sinks* (gain-side). Sinks drive the inter-group coupling.

6.4.4 Specific hermetry-form solutions

Remark 6.7 (Form-by-form solutions). For each of a, b, c, d the construction of Theorems 6.15–6.16 proceeds by:

1. writing down $\hat{\gamma}_x$ from (6.3)–(6.6);
2. identifying the non-vanishing condensers via (6.19);
3. solving (6.16) in the second validity regime;
4. reading off the coupling-group schema and the corresponding spectrum from (6.22).

For form d (total enneametric) all 81 condensers participate; for form a (self-condensation) only a single condensor survives. The detailed component-wise expressions for each form are presented in the source and will be invoked in subsequent chapters as needed.

6.5 Field Activation and Condensor Flux

6.5.1 Consistency of the proto-field/activator decomposition

Imposing the constancy ${}^2\tilde{\gamma}_{(\kappa\lambda)-}; n = \text{const}$ on every kernel and writing ${}^2\tilde{\kappa}_{(\lambda)\pm} \sim f_{\lambda}^{\pm 1}$ with $f_{\lambda} = f(\mu_{\lambda})$ depending only on the hermetric lattice selectors of λ :

Theorem 6.17 (Activator constancy condition). *The hermetric functions f_{κ}, f_{λ} of the lattice kernels satisfy*

$$\boxed{\frac{f_{\kappa}^2 + f_{\lambda}^2}{f_{\kappa} f_{\lambda}} = \beta_{\kappa\lambda} = \text{const}, \quad \beta_{\kappa\lambda} \neq 2 \text{ for } \kappa \neq \lambda, \quad \beta_{\lambda\lambda} = 2,} \quad (6.24)$$

together with

$$f_{\kappa} f_{\lambda} \sim |(e^{\lambda_l \mu} - E)^{\alpha_l} \delta_{kl}|_{q'}, \quad \mu = \mu_{\kappa} + \mu_{\lambda}. \quad (6.25)$$

The constants $\beta_{\kappa\lambda}$ govern the off-diagonal inter-structure couplings; the diagonal case $\beta_{\lambda\lambda} = 2$ is the partial-structure self-coupling normalisation.

6.5.2 Coordinate dependence of the condensor schemata

Theorem 6.18 (Invariance of the condensor schema). *The schema of fundamental condensers of a multi-structure-unit hermetry form (i.e. b, c, d) is an invariant of every admissible coordinate transformation. There is no global frame on which more than one structure unit can be simultaneously transformed away.*

Only form a admits a global frame $C_{(11)}$ in which all its condensor entries vanish; the resulting “free fall” is the geodesic motion of pure gravity, and the apparent forces of gravity are frame-dependent pseudo-forces.

6.5.3 The structure compressor as condensation potential

Definition 6.8 (Synmetronic structure compressor). For each fundamental condensor of a hermetry form there is an attached *structure compressor*

$${}^4\tilde{\zeta}_{(\kappa\lambda)}^{(\mu\nu)} = K_{(\kappa\lambda)}^{(\mu\nu)}; \left[\frac{\kappa\lambda}{\mu\nu} \right]_+ \quad (6.26)$$

playing the role of the discrete Riemann curvature of that partial structure. The geodesic of the partial structure satisfies

$$\ddot{x}^i_{(\kappa\lambda)} \left(\begin{smallmatrix} \mu\nu \\ \kappa\lambda \end{smallmatrix} \right) = - [i]_{(k\lambda)+}^{(\mu\nu)} \dot{x}^k \dot{x}^l. \quad (6.27)$$

6.5.4 Variational principle: the condensor flux

Theorem 6.19 (Condensor-flux variational principle). *The structure compressor of every fundamental condensor satisfies a universal minimum-energy variational principle:*

$$\boxed{\delta \int_{n_1}^{n_2} {}^4\tilde{\zeta}_{(\kappa\lambda)}^{(\mu\nu)}; n \, \partial n = {}^4\bar{0}.} \quad (6.28)$$

With increasing metron count the compressor drives every partial structure toward its lowest available compression niveau; the trajectory of the partial structure in R_6 is the metron-quantised analogue of the geodesic of a Riemannian connection.

6.5.5 Universal conservation law in the compositional field

For the compositional field $\widehat{\{\}}$ specifically, the structure compressor obeys an additional, exact conservation law:

Theorem 6.20 (Compositional conservation). *In the compositional field,*

$$\boxed{\mathrm{sp} \left(\begin{smallmatrix} & & \\ & & \\ & & \end{smallmatrix} \right)_{(-)}^{(6,6)} ; \left(\left(\begin{smallmatrix} & & \\ & & \\ & & \end{smallmatrix} \right) - \frac{1}{2} {}^2\bar{\gamma} \mathrm{sp} \left(\begin{smallmatrix} & & \\ & & \\ & & \end{smallmatrix} \right) \right) ; \mathrm{sp} {}^4\bar{\zeta} = \bar{0},} \quad (6.29)$$

which expresses the divergence-freedom of the composite energy density in the cellular calculus. This is the discrete analogue of the contracted Bianchi identity ([Wikipedia: Bianchi identities](#)); the conservation is exact in the composite field, even when the partial structures individually carry non-conserving condensor fluxes.

6.5.6 Transmutation as activator activation

Remark 6.9 (Closure). Every *transmutation* of a composite hermetry form $a \rightarrow b \rightarrow c \rightarrow d$ proceeds via the induction of a compensating spin field (Theorem 6.11), which in turn activates the proto-field of the partner partial structure (Corollary 6.4), whose curvature ${}^4\zeta_{(\kappa\lambda)}^{(\mu\nu)}$ is then driven downward by the condensor flux (6.28), while the composite energy density remains conserved by (6.29). The mechanism by which the discrete particle spectrum acquires its empirical pattern of correlations, ground states, resonances, and decay channels is controlled in full generality by this chain.

The chapters that follow apply this machinery to the construction of the discrete particle spectrum, the isospin families of ground states, the resonance spectra, and the global symmetry algebra of the unified theory.

Summary of notation introduced in this chapter

Symbol	Object
${}^2\bar{\kappa}_{(1)}, {}^2\bar{\kappa}_{(2)}, {}^2\bar{\kappa}_{(3)}$	Three structure-unit lattice kernels
$\hat{\gamma} = ({}^2\bar{\gamma}_{(\mu\nu)})_3$	Polymetric correlator
$\hat{\gamma}_{(H)}, \hat{\gamma}_{(A)}$	Hermetric / antihermetric blocks of $\hat{\gamma}$
${}^2\bar{\gamma}_{(\mu\nu)+} = {}^2\bar{\gamma}_{(\mu\nu)+}$	Proto-field
${}^2\bar{\gamma}_{(\mu\nu)-} = {}^2\bar{\gamma}_{(\mu\nu)-}$	Field activator
${}^2\bar{P}_{(\mu\nu)} = (1/\tau) \text{ROT}_{(x)} \bar{\phi}_{(\mu\nu)}$	Discrete spin selector
$S(\mu)$	Sieve operator (antihermetises unit μ)
$\hat{\gamma}_a, \hat{\gamma}_b, \hat{\gamma}_c, \hat{\gamma}_d$	Hermetry-form correlators
Partial fundamental condensor	with basis $(\mu\nu)$, contra $(\kappa\lambda)$
$L_{(\mu\nu)}^{(\kappa\lambda)}$	Synmetronic world selector
$\bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)}$	Partial eigenvalue spectrum
${}^2\bar{Q}_{(\mu\nu)}^{(\kappa\lambda)}$	Inter-structure coupling cochain
$K_{(\kappa\lambda)}^{(\mu\nu)}$	Synmetronic Bianchi raising selector
$D_{(\mu\nu)}^{(\kappa\lambda)}$	Quadratic coupling selector
$Q_{(\mu\nu)}^{(\kappa\lambda)}$	Coupling selector (6.21)
${}_{(\mu\nu)}^{(\kappa\lambda)}$	Coupling-group symbol
$\beta_{\kappa\lambda}$	Activator-consistency constant
$4\bar{\zeta}_{(\kappa\lambda)}^{(\mu\nu)}$	Partial structure compressor
$Z = \omega^2 q(2\omega - q)$	Spectral selection count

7

Correlation and Correspondence

Scope of the chapter. With the synmetronic equation of Chapter 6 in hand, the discrete metric condensation stages $\bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)}$ of any hermetry form can be decomposed into their internal coupling structure. The present chapter develops the dynamical content of those couplings: the *flux aggregates* that link condensor extrema in cycles, the *prototropic conjunctives* that classify their connectivity, the *straton* and *conjunctor spins* attached to each cycle, the *mirror antistructure* that completes the four-spacetime to its full R_6 form, and the *correspondence fields* that arise when two condensations interact.

The chapter closes with a structural derivation that fixes the spatial dimension of the universe at $p = 3$: the only consistent extension of the synmetronic ansatz down to its most primitive prototropic element — the *ur-anisotropy* U , a single relative spin reversal of a single metron — forces the state space of U to be a two-dimensional complex vector space whose $SU(2)$ rotation group is isomorphic to $SO(3)$ of a real three-dimensional compact manifold, identified with the physical R_3 .

Five sections develop the construction. Section 7.1 catalogues the cyclic flux aggregates of the coupling structure and derives the universal *inertia principle* $\bar{\lambda} \perp Y$ that constrains them. Section 7.2 introduces the protosimplexes $(\pm p)_x$ and their prototropic conjunctives. Section 7.3 defines the conjunctor and straton spins and classifies all hermetry-form solutions into *tensor terms* (integer spin, no exclusion) and *spinor terms* (half-integer spin, exclusion principle). Section 7.4 establishes the R_4^- antiworld and the mirror antistructures. Section 7.5 develops the correspondence schema and closes with the seventeen-step structural hierarchy that fixes $p = 3$.

7.1 Flux Aggregates and the Inertia Principle

7.1.1 Cyclic flux aggregates

By the coupling-extremum theorem of Section 6.4, every coupling group $\binom{\kappa\lambda}{\mu\nu}$ is a stationary point of an internal cyclic condensor flux that exchanges occupancies between coupling extrema. We classify these cyclic aggregates by the number ν of distinct *coupling classes* they engage and by the cyclic length ω that returns the flux to its starting condition.

Theorem 7.1 (Cyclic-aggregate inventory). *The cyclic flux aggregates*

of the coupling structure are exhausted by the following counts:

ν	degeneracy	aggregate count
1	double	4 aggregates (j)1, $1 \leq j \leq 4$
2	single	$4 \times 3 = 12$ aggregates $(\cdot, \cdot)3$
3	single or regular	4 degenerate($\omega=2$) + 4 regular($\omega=4$), $v! = 6$ isomers each
4	regular only	15 types, $v! = 24$ isomers each, first stereoisomerism
5	regular only	6 types, $v! = 120$ isomers each, variable ω
6	regular only	1 type, 720 isomers.

(7.1)

Remark 7.1 (Enantiostereoisomers and structural isomerism). At $\nu \geq 4$ the aggregates first acquire genuine structural isomers (analogous to optical antipodes in chemistry), distinct under spatial reflection in R_6 . Each structural isomer carries a paired *enantiostereoisomeric* mirror image — a right- and a left-handed cyclic orientation that play distinct physical roles. For $\nu \leq 3$ only spin isomerism is present.

7.1.2 The inertia principle

Substituting the synmetronic geodesic $\ddot{C}^i + [^i_{kl}]; n \dot{C}^k \dot{C}^l = 0$ into the component form of the world selector $L\{\widehat{\cdot}\} = {}^4\bar{0}$ and contracting with $\dot{C}^k \dot{C}^l \dot{C}^m$ forces an orthogonality between the eigenvalue spectrum and the six-velocity. The same construction goes through signature by signature for the synmetronic problem:

Theorem 7.2 (Universal inertia principle). *For both the composite and the partial structures,*

$$\bar{\lambda} \perp \mathbf{Y}(q), \quad \bar{\lambda}_{(\kappa\lambda)}^{(\kappa\lambda)} \perp \mathbf{Y}(z), \quad (7.2)$$

with $\mathbf{Y}(q) = \sum_{i=1}^q \dot{x}_i$ the world velocity in the q -dim hermetric subspace, $z = z(\kappa\lambda\mu\nu) \leq q \leq 6$ the hermetric dimension of the $(\kappa\lambda, \mu\nu)$ -condensor, and $\mathbf{Y}(6) = v + iw$ with $w^2 = c^2 + \dot{\epsilon}^2 + \dot{\eta}^2$.

Corollary 7.3 (Genesis of inertia). *The inertia of every material structure is the reaction of the orthogonality (7.2) to a perturbation $v \neq 0$: the eigenvalue vector $\bar{\lambda}$ resists any motion that would violate its orthogonality to the local six-velocity. The phenomenological inertial mass is the proportionality constant of this reaction.*

7.1.3 The space-condensation class [3]

A direct consequence of (7.2) is that only those condensor entries whose contra-signature is the purely real-space $(3, 3, 3, 3)$ can avoid forming a discrete spectrum. Define

$$[3] = \left\{ \left[\begin{smallmatrix} 3 \\ 33 \end{smallmatrix} \right]_{(,)}^{(3)} \left[\begin{smallmatrix} 3 \\ 33 \end{smallmatrix} \right]_{(33)}^{(33)}, \dots \right\}. \quad (7.3)$$

Theorem 7.4 (Space-condensation class). *The condensor entries in [3] depend only on ${}^2\bar{\kappa}_{(3)}$, hence only on the real R_3 coordinates. They produce*

a continuous spectrum that envelops the discrete imaginary spectra. In the infinitesimal limit $\tau \rightarrow 0$ they collapse to a Yukawa-like near-field

$$[3]; n \rightarrow \psi(r) \sim e^{-\alpha r}, \quad \alpha > 0. \quad (7.4)$$

Thus [3] is responsible for the ponderability of the c and d forms (it provides their cellular near field), while the imaginary imponderable forms a, b — which lack ${}^2\bar{\kappa}_{(3)}$ — admit only stage-bearing condensations.

7.1.4 Flux velocity and the wavelength-mass relation

By Theorem 7.2 together with the imaginary-kernel origin of every condensation stage, the cyclic flux velocity inherits the imaginary component of the six-velocity:

Theorem 7.5 (Flux velocity). *The cyclic flux velocity of any aggregate satisfies*

$$w_f = w \geq c, \quad m\lambda = \mu\lambda_0 = \text{const}, \quad (7.5)$$

with the cyclic diameter λ identified as the Compton wavelength of the composite quantum. For imaginary forms (a, b) , $\lambda_{(a,b)}/\lambda_0 \sim E_{(a,b)}^{-1}$. For complex forms (c, d) ,

$$2n(\lambda_{(c,d)}/\lambda_0)^4 = (2n-1)^2(1+4(q/\pi)^4). \quad (7.6)$$

7.2 Protosimplexes and Prototropic Conjunctives

7.2.1 Fluktons and screen fields

Definition 7.2 (Flukton, screen field, protosimplex). A *flukton* $(-p)_x$ is the active cyclic condensor flux of class p in hermetry form x . Its dynamic stability requires a surrounding *screen field* $(+p)_x$ — the static, kondensor-flux-free envelope that completes the cycle into a closed protostructure. Together they form the *protosimplex*

$$(\pm p)_x := (-p)_x \cup (+p)_x. \quad (7.7)$$

Theorem 7.6 (Strata and pseudo-strata). *The screen field $(+7)_x$ — the seventh-class envelope — exists only for the complex hermetry forms $x \in \{c, d\}$, and its R_3 -projection constitutes the straton, a discrete real R_3 -region that carries the ponderability of the quantum. The imaginary forms a, b produce only pseudo-strata: R_3 regions of influence without a $(+7)_x$ envelope.*

7.2.2 Prototropic conjunctives

Definition 7.3 (Prototropic conjunctive). The internal correlation $(\pm p)_X \rightarrow (\pm q)_Y$ between two protosimplexes acting through the structure unit $\mu \leq 3$ is the *prototropic conjunctive*

$$(\pm p)_X - (\mu) - (\pm q)_Y. \quad (7.8)$$

The hyphens indicate the active condensor-flux channel; the symbol in parentheses identifies the lattice kernel ${}^2\bar{\kappa}_{(\mu)}$ that mediates the conjunctive.

Remark 7.4 (Layered notation). For directional conjunctives we write $(\pm p)_X - (\mu) - (\pm q)_Y$ (flow toward Y) and $(\pm p)_X - (\mu) - (\pm q)_Y$ (flow toward X); their superposition is the simultaneous conjunctive $(\pm p)_X - (\hat{\mu}) - (\pm q)_Y$.

7.3 Conjunctor and Straton Spin

7.3.1 Conjunctor spin

Definition 7.5 (Conjunctor spin). Each prototropic conjunctive carries an attached spin selector

$$p_x(\mu)^q = \pm \bar{s}_\mu \frac{\hbar}{2} \underline{m}_x^{(pq)}, \quad |p_x(\mu)^q| = \bar{s}_\mu |p_x(\mu)^q|, \quad (7.9)$$

in which $\bar{s}_\mu$ is the unit spin vector of the kernel μ and $\underline{m}_x^{(pq)}$ is the integer quantum count.

7.3.2 Straton spin

Theorem 7.7 (Straton spin). The full straton spin of a hermetry-form composite quantum is the sum over all conjunctor spins,

$$\bar{s}_x = \sum_{\mu, p, q} p_x(\mu)^q = \pm \bar{s}_0 \hbar m_x / 2, \quad \bar{s}_0 = \sum_{\mu} \bar{s}_\mu, \quad m_x = \sum_{p, q} \underline{m}_x^{(pq)}. \quad (7.10)$$

For a running metron count v , the measurable spin and its selector are

$$\bar{s}(x) = \pm \hbar \bar{\sigma}(x), \quad 2\bar{\sigma}(x) = \bar{s}_0 m_x; v, \quad \bar{\sigma} \neq \bar{\sigma}^*. \quad (7.11)$$

7.3.3 Trans-spin and space-spin decomposition

In the imaginary subspace $V_3(x_4, x_5, x_6) \perp R_3$ the straton spin decomposes as

$$\bar{\sigma}(x) = i(\bar{e}_t s + \bar{e}_r J P_a), \quad 2s = P \geq 0, \quad 2J = Q \geq 0, \quad P_a = \alpha_p (-1)^J, \quad \alpha_p = \pm 1. \quad (7.12)$$

Theorem 7.8 (Tensor / spinor dichotomy). Depending on the parity of $Q = 2J$ the hermetry-form composite is either:

$$\begin{aligned} Q = 2n : \quad & \sigma_r = \pm i J_T, \quad J_T = n, \quad 0 \leq n \leq 6 \quad (\text{tensor term}), \\ Q = 2n + 1 : \quad & \sigma_r = \pm J_S, \quad 2J_S = 2n + 1, \quad n \geq 0 \quad (\text{spinor term}). \end{aligned} \quad (7.13)$$

Tensor terms carry no R_3 -volume exclusion (their imaginary σ_r disjoins them from the real R_3). Spinor terms do carry exclusion (their real σ_r is interwoven with R_3).

Corollary 7.9 (Graviton and photon assignments). For the imaginary forms restricted to their R_4 -projectable subset, the spin assignments are

$$P(a_g) = P(b) = 0, \quad J_T(a_g) = 2, \quad J_T(b) = 1. \quad (7.14)$$

The graviton (a_g) is therefore a spin-2 tensor term; the photon (b) is a spin-1 tensor term. Both carry zero singlet count P .

Remark 7.6 (Subconstituent counting). For complex forms a coupling parameter k defines the maximum singlet count via

$$0 \leq P \leq G = k + 1. \quad (7.15)$$

The integer G counts the metaphorical *quasi-corpuseular subconstituents* of the straton (+7); these subconstituents cannot appear as free corpuscles because they exist only as symmetry components of the screen field. All members of a spin-isomorphic multiplet share the same k and so the same G .

7.4 Antisttructures

7.4.1 The antiworld R_4^-

Theorem 7.10 (Antisttructure existence). *The orthogonality $\bar{\lambda} \perp \bar{x}_4$ of the inertia principle holds with $\cos \alpha_4 \neq 0$. Both signs are geometrically admissible:*

- $\cos \alpha_4 > 0$: the world R_4^+ with parallel spaces R_3^{+k} , $k \geq 0$, $R_3^{+0} \equiv R_3$;
- $\cos \alpha_4 < 0$: the antiworld R_4^- with parallel anti-spaces R_3^{-k} .

Both half-worlds carry mirror-symmetric spin systems and obey

$$\cos(\mathbf{Y}^+, \mathbf{Y}^-) = -1, \quad \cos(\bar{\lambda}^+, \bar{\lambda}^-) = -1, \quad (7.16)$$

so that $\bar{\lambda}^\pm \perp \mathbf{Y}^\pm$ holds in both half-worlds.

7.4.2 Antisttructures in R_4^+

Theorem 7.11 (Antisttructures during transmutation). *Antimatter as a macroscopic body is not possible in R_4^+ — its proper home is R_4^- . However, during transmutation of an elementary $\mathcal{M}_q$ in R_4^+ , the conservation of angular momentum demands that mirror-spin anti-corpuseles appear in R_3^{+0} as transient products:*

$$\boxed{\bar{s}_x^- := \bar{s}_x(R_4^-) \text{ appears in } R_3^{+0} \text{ at transmutation events of } \bar{s}_x^+,} \quad (7.17)$$

i.e. standard particles in R_4^+ are accompanied by their R_4^- -rooted anti-particles in scattering and decay channels. The protosimplex of the anti-corpusele is the enantiostereoisomer $(\pm p)_x^-$ of $(\pm p)_x^+$, with all conjunctor spins, conjunctor signs, and electric charges reversed.

Corollary 7.12 (Antisttructure ponderability). *Since $(\pm p)_x^-$ is the enantiostereoisomer of $(\pm p)_x^+$, the imaginary hermetry forms $x \in \{a, b\}$ produce imponderable antisttructures and the complex forms $x \in \{c, d\}$ produce ponderable antisttructures with charge mirrored.*

7.5 Correspondence Fields and the Universal Hierarchy

7.5.1 Two classes of correspondence

Definition 7.7 (Rk and Sk). An interaction between two composite quanta X and Y via their condensation eigenvalues $\bar{\lambda}_X, \bar{\lambda}_Y$ is

- a *Raumspin* correspondence Rk if it proceeds through the spin vectors $\bar{s}_X, \bar{s}_Y$;
- a *Strukturkorrespondenz* Sk if it proceeds through the coupling structure of X, Y irrespective of spin orientation.

Theorem 7.13 (Rk existence). *The Raumspin correspondence exists iff*

$$\boxed{J_X > 0, \quad J_Y > 0, \quad \cos(\bar{s}_X, \bar{s}_Y) \approx \pm 1.} \quad (7.18)$$

7.5.2 The four-step Sk schema

Theorem 7.14 (Sk unfolding). *A general structure correspondence between X and Y in the kernel channel μ proceeds through four stages:*

$$(+p)_X - [\hat{\mu}] - (+q)_Y \rightarrow (\pm p)_X - (\mu) - (\pm q)_Y \quad (\text{screen-field corresp.,} \quad (7.19)$$

then $X \rightarrow Y$ flux),

$$(\pm p)_X - (\mu) - (\pm q)_Y \rightarrow (\pm p)_X - (\hat{\mu}) - (\pm q)_Y \quad (Y \rightarrow X \text{ flux, then bidirectional}).$$

The final state is a new coupled composite $\Lambda(X, Y)$ whose internal cycle replaces the previously independent cycles of X and Y .

7.5.3 Thirteen correspondence classes

Theorem 7.15 (Correspondence-class count). *Classified by their dominant kernel channel and the role of screen fields versus fluktons:*

$$\boxed{\begin{array}{l} \text{weak : } Rk \cup \{6 \text{ screen-field correspondences } (+p)\text{-corresp.}\}, \\ \text{strong : } \{6 \text{ flukton correspondences } (-p)\text{-corresp.}\}, \end{array}} \quad (7.20)$$

giving $1 + 6 + 6 = 13$ correspondence classes. The seven weak classes have $\ddot{x}^i$ contributions that are sub-leading; the six strong flux classes carry the dominant ponderomotive $\ddot{x}^i$.

Corollary 7.16 (Gauge invariance). *Gauge invariance attaches to the flukton correspondences and not to the screen-field correspondences. The fact that [Wikipedia: Maxwell's equations](#) are gauge-invariant but Newton's φ_n is not therefore follows from the structural hierarchy: electromagnetism is a flukton correspondence, gravitation a screen-field correspondence.*

7.5.4 Counting the correspondence sources

Theorem 7.17 (Correspondence-source count). *For hermetry form x with $n = n(\hat{\gamma}_x)$ entries of the correlator matrix, the number of structural correspondence sources is*

$$\boxed{Z(x) = n(n-1),} \quad (7.21)$$

n being the non-vanishing entry count of $\hat{\gamma}_x$. From Theorem 6.2:

$$Z(a) = 2, \quad Z(b) = Z(c) = 30, \quad Z(d) = 72. \quad (7.22)$$

7.5.5 Hierarchical structure of the universe

The symmetronic principles assemble into a 17-step structural hierarchy of the universe — from cosmic super-clusters at the top down to the irreducible *ur-anisotropy* U at the bottom:

Theorem 7.18 (Universal hierarchy). *The structural hierarchy is:*

1. *Macroscopic attraction by the modified Newton field $r q e^{-q} \sim (1 - r/\zeta)^2$ governs all matter at $r \gg \zeta$ or $r \geq \zeta$; spiral-nebula systems are the last attractively bound macroscopic units, beyond which matter is chaotic.*
2. *At $0 \ll r < \zeta$, spiral-nebula systems are attractive groupings of individual galaxies.*
3. *A spiral galaxy at $0 \ll r \ll \zeta$ is a stable attractor of its star clouds and dark matter.*
4. *Star clouds resolve into individual stars supporting stable planetary systems.*
5. *Each cosmic object is a heteromorphic matter assembly serving simultaneously as source of all distance-mediated correspondences.*
6. *All matter resolves at the molecular and atomic level into intermolecular attractors.*
7. *Molecular structure is atomistic and obeys valence rules.*
8. *Each atom is Z shell electrons around a Z -positive nuclide.*
9. *Nuclides and electron shells alike are built only from c, d hermetry forms, joined by the strong nuclear correspondence (p, n in the nucleus) and the weak electromagnetic correspondence (electron shell).*
10. *All micro- and macro-states correspond through imaginary hermetries a, b ; their elements are always ponderable c, d condensations.*
11. *By the conjunctive law $-(\mu)-$, the elementary hermetry forms are themselves composed from prior forms — the protosimplexes.*
12. *Internal structure of $\mathcal{M}_q$ as conjunctive relations of protosimplexes is a consequence of the symmetronic equation (6.16) which exhibits them as prototropic aggregates.*
13. *Every prototrope is either a real flukton or a degenerate screen field; protosimplexes are fluktons enveloped by static screen fields.*
14. *Every prototrope is, at the cellular level, a metronic condensation state of R_6 or a subspace.*
15. *Every condensation state is built from elementary metronic condensation stages of the symmetronic equation.*
16. *The pre-form of every condensation stage is a field activation on a fully isotropic empty patch of the metronic world tensor manifold — a spin anisotropy in the absence of any condensation.*
17. *The irreducible element of every spin anisotropy is the *ur-anisotropy* U : a single relative spin reversal of one metron against the isotropic empty tensor manifold.*

7.5.6 The dimension theorem from U

Theorem 7.19 (Dimensional uniqueness). *The state space of U is a two-dimensional complex vector space (the alternative $a_+ =$ exists vs. $a_- =$ does not exist is the only property of U). The natural automorphism group of this state space is $SU(2) \cong SO(3)$ of a compact real three-dimensional manifold. Therefore the only image space of U is a compact real three-dimensional space; identifying it with the semantic unit $s_{(1)}$ of the world manifold,*

$$\boxed{p = 3, \quad U \text{ and } R_3 \text{ are mutually constitutive.}} \quad (7.23)$$

The earlier dimensional bound $p \leq 3$ of Theorem 2.19 is hereby sharpened to the exact equality.

Remark 7.8 (The three empirical principles as consequences). The empirical conservation laws 1.1, the variational structure 1.2, and the action quantum 1.3 of Chapter 1 all follow as consequences of the world selector $L\{\widehat{\cdot}\} = {}^4\bar{0}$ with $\tau > 0$ and the resulting prototropic structure:

- Energy and electric-charge conservation: the conservation of the discrete kompressor / kondensor pattern of R_6 under all transmutations (the compressor isostasy law).
- Angular-momentum conservation: the constancy of every field activator and hence of every spin field.
- Entropy increase / minimum energy: the variational principle of Theorem 6.19 applied to the integral kompressor.
- Quantisation $E = nh\nu$: the discrete cyclic flux aggregates of Section 7.1.
- Three-dimensional physical space: Theorem 7.19 just established.

What had to be *postulated* in Chapter 1 is now *derived* from the cellular discrete structure of R_6 .

Summary of notation introduced in this chapter

Symbol	Object
$(-p)_x$	Flukton: active cyclic condensor flux
$(+p)_x$	Screen field: static envelope
$(\pm p)_x$	Protosimplex (flukton + screen)
$(+7) = (+7)_x$	Straton (ponderability envelope of c, d)
$-(\mu)-$	Prototropic conjunctive in kernel μ
${}_x^p(\mu)^q$	Conjunctor spin
$\bar{s}_x = \pm \bar{s}_0 \hbar m_x / 2$	Straton spin
$\bar{\sigma}(x) = i(\bar{e}_t s + \bar{e}_r J P_a)$	Trans + space spin decomposition
P, Q	Singlet and angular-momentum quantum numbers
$P_a = \alpha_p (-1)^J$	Parity
J_T, J_S	Tensor / spinor space-spin
$k, G = k + 1$	Coupling parameter, max subconstituent count
$[3]$	Real-space-only condensation class
R_4^-, R_3^{-k}	Antiworld and antipararäume
$\bar{s}_x^\pm$	Spin of normal / mirror corpuscle
Rk, Sk	Spin vs. structure correspondence
$Z(x) = n(n-1)$	Correspondence-source count
U	Ur-anisotropy of a single metron
$Y(q)$	World velocity of dimension q subspace